Jumping to Conclusions: Polymarket Price Jumps Analysis

An exploratory analysis of 766 million Polymarket trades, November 2022 to May 2026

Code and plots: github.com/CrabGuy/PolymarketPriceJumpsAnalysis

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1. In short

On Polymarket, people bet on events by buying shares that pay $1 if the event happens, so a share's price is the market's probability for the event. We looked at every sharp price jump in 766 million trades and asked one question: **judging by how the market finally resolved, how much of the jump was justified?**

Just under two thirds. The rest, a little over a third, is an **overshoot**: noise that the market later takes back. This holds in every year and every category.

What tells real jumps from noise is **how many different traders made them**:

- A jump made by one trader goes about 40% too far, and the next trades, minutes later, undo most of it.
- A jump made by a crowd (five traders or more) goes about a seventh too far, and that part lingers for hours.
- Lone pushes are common where the order book is thin, so that a few dollars can move the price.

Once the bid-ask spread is paid, betting against lone pushes loses money. Betting against crowd jumps, and jumps with $200 or more behind them, leaves a small edge before fees.

2. The data and two ideas

The data

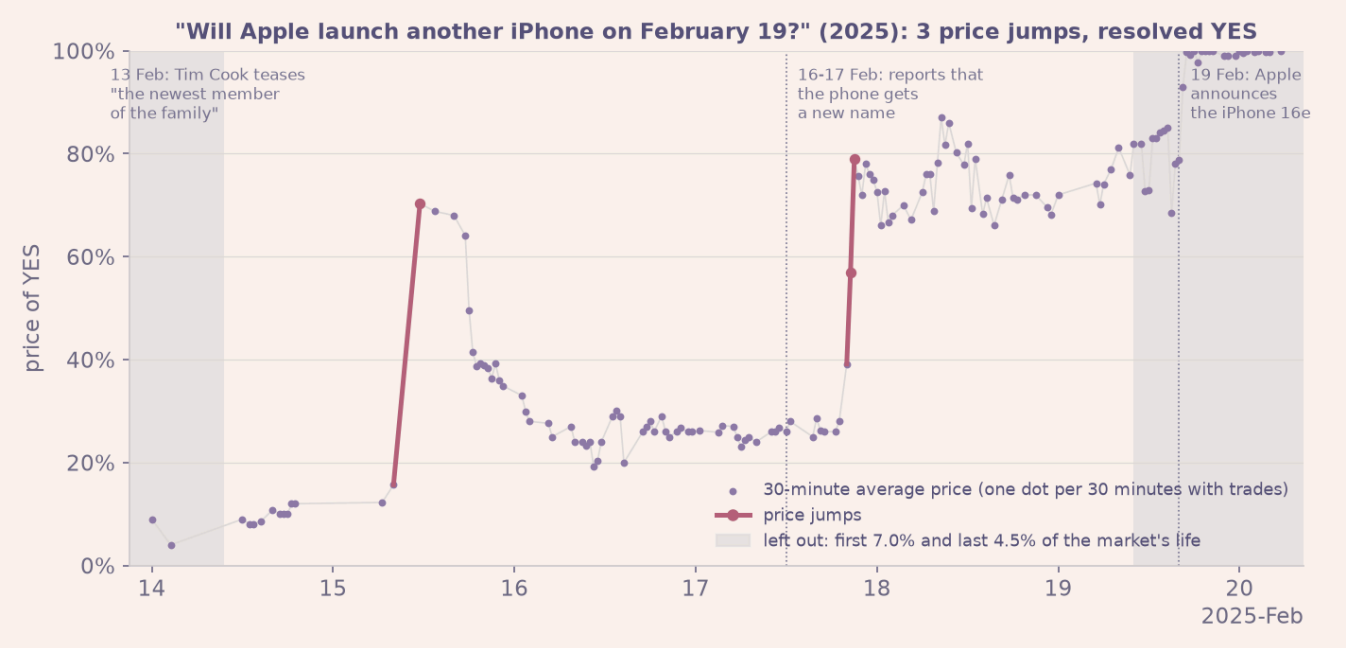
We used every Polymarket trade from November 2022 to May 2026: 766 million trades in about one million markets, each with a time, a price, a dollar amount and the two wallets involved. The heavy work ran on a server; small summary tables were analysed on a PC.



Most of the report covers **event markets**: elections, sports, economy, culture, and so on. That is about 430,000 markets. The recurring crypto "up or down" markets ("Will bitcoin be up in the next 15 minutes?") behave differently and are covered in F8.

Idea 1: a price jump

Here is a real market: *"Will Apple launch another iPhone on February 19?"* (2025), a small market with 599 trades. On 13 February Tim Cook announced "the newest member of the family" for 19 February, and Polymarket opened one market per possible product. A separate market covered an "iPhone SE", the name everyone expected, so this one was in effect a bet on a different name. It traded near 10% for two days. On Saturday 15 February one trader pushed it to 70% with no news we could find, and other traders sold it back to 25% by Monday. On 16-17 February reports spread (Bloomberg's Mark Gurman first) that the phone would get a new name; on the afternoon of the 17th (US time) several traders lifted the price from 28% to about 70%, and it held. On 19 February Apple announced the **iPhone 16e**: the market resolved **YES** (and the iPhone SE market NO).



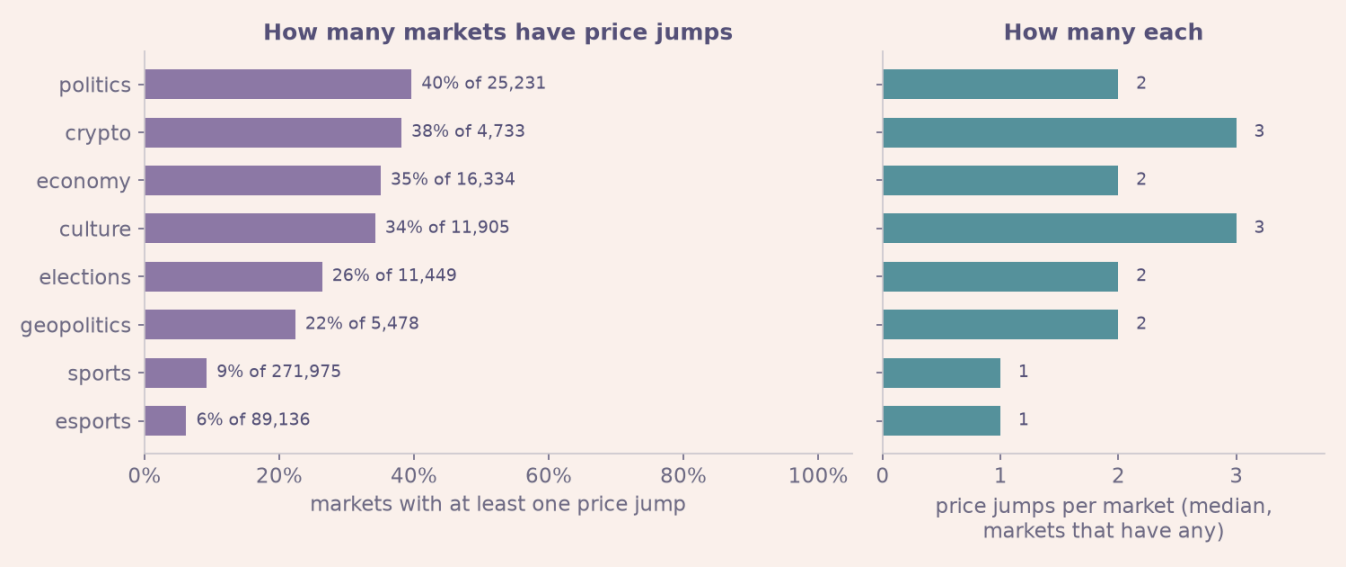
The trades are grouped into 30-minute slots, and each slot gets its average price: each dot is one slot with trades. A **price jump** (red) is a slot whose price is at least **15 percentage points** away from the previous slot with trades.

Two stretches of each market's life are left out (grey). The life runs from the first trade to the moment the price leaves 3-97% for good (or to the last trade, if it never does):

- **the first 7%**, while the price is still being found;
- **the last 4.5%**, the run-up to 0% or 100% that every market makes, which would only measure the obvious.

Because both scale with the market, a three-hour game loses about 13 minutes at the start and 8 at the end, and a year-long election market about four weeks and two weeks. In this market that leaves three jumps: the lone push on the 15th, and the rise on the 17th in two steps. The announcement itself falls in the last 4.5%. Every number in this definition (30 minutes, 15 points, 7%, 4.5%, 3%) was set by the elbow method, below.

How common are price jumps? 40% of political markets have some; most sports and esports markets, which are short and many, have none:



Two facts about these jumps matter for everything that follows.

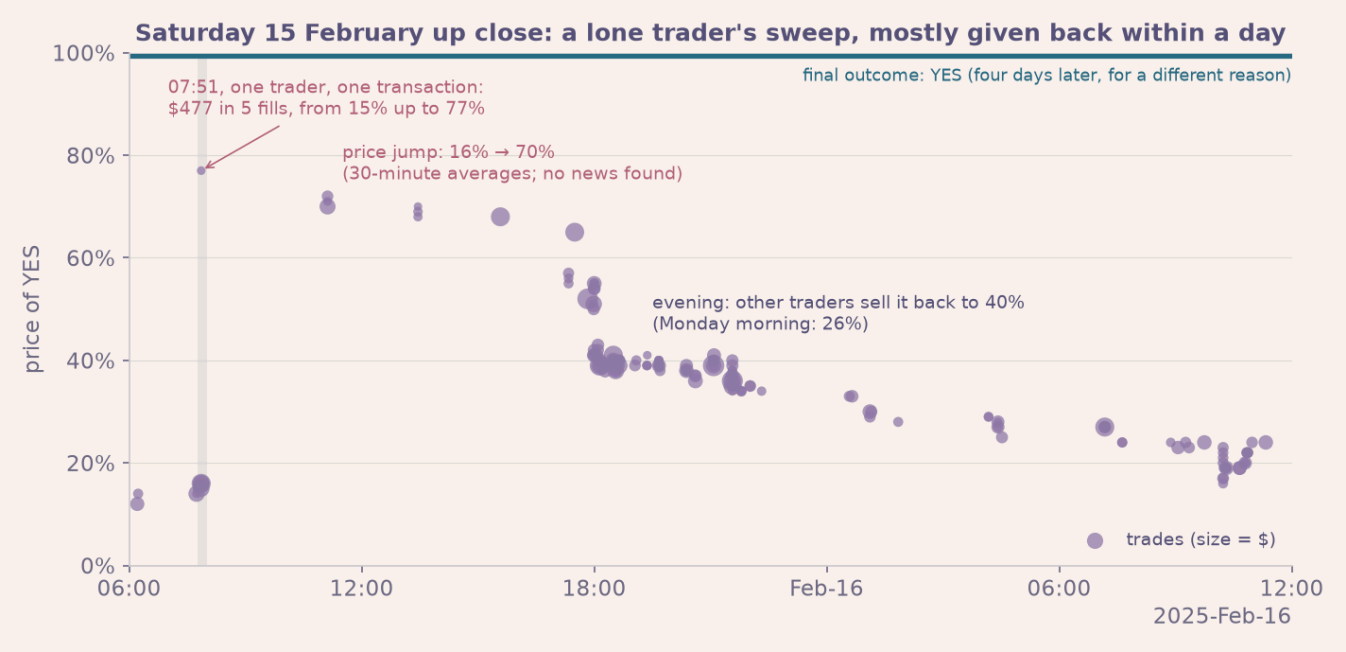
They involve little money. The median price jump has $10 traded in its 30 minutes, and 78% have less than $100. A price can jump on a few dollars because, in most markets, few orders sit near the current price. A small order runs through them and reaches prices far away.

Most are made by a single trader "taking liquidity". Some traders post an order and wait ("I'll sell at 30%"): they are the *makers*. Others accept a posted order immediately: they are the *takers*, and only takers move the price. So for each price jump we count the different wallets that took liquidity in its 30 minutes. In 65% of price jumps there is just one, and that trader spent a median of $4.

Idea 2: overshoot

Zoom in on Saturday 15 February. At 07:51, in one transaction, a single trader bought $477 of YES in five fills: four at 15-16% and a last one at 77%, which left the price there. The next trades were at about 70%, so the 30-minute average jumped from 16% to 70%. No news came out; by the evening other traders had sold it

back to 40%, and by Monday morning it was at 26%. Most of the jump was given back. (The market still resolved YES, four days later, for a reason this trader could not have known on Saturday.)



The part of a jump that turns out to be unjustified is the **overshoot**. One market can't measure it, but thousands can:

1. Take all jumps that ended at, say, 60%.
2. If those prices were right, YES should win 60% of the time.
3. If YES wins only 53% of the time, the prices were 7 points too high.
4. Against a 20-point jump (40% → 60%), that is an overshoot of 7 / 20 = **35% of the jump**.

Prices are slightly off even at quiet moments (YES tends to be a little overpriced), so that usual error is measured at random moments and subtracted first. An overshoot of 0% means the jumps were exactly right; a negative one means the price didn't go far enough.

In formulas, for a jump j from price p_{j^-} to p_j, with outcome O_j (1 if YES won, 0 if NO won), direction d_j (+1 up, −1 down) and $b(p)$ the usual error at price p:

$$o_j = -\left[(O_j - p_j) - b(p_j)\right] \cdot d_j, \qquad \text{overshoot} = \frac{\sum_j o_j}{\sum_j |p_j - p_{j^-}|}$$

Uncertainty ranges come from resampling whole markets. Every formula used in the report is listed in section 6.1.

A note on averaging. Each 30-minute price is volume-weighted: a trade counts in proportion to its dollars. Weighting only acts *within* a slot, so a slot traded by one wallet simply gets that wallet's average fill. Re-measuring with plain averages, every trade counting equally, gives the same conclusions, with crowd overshoots a few points lower.

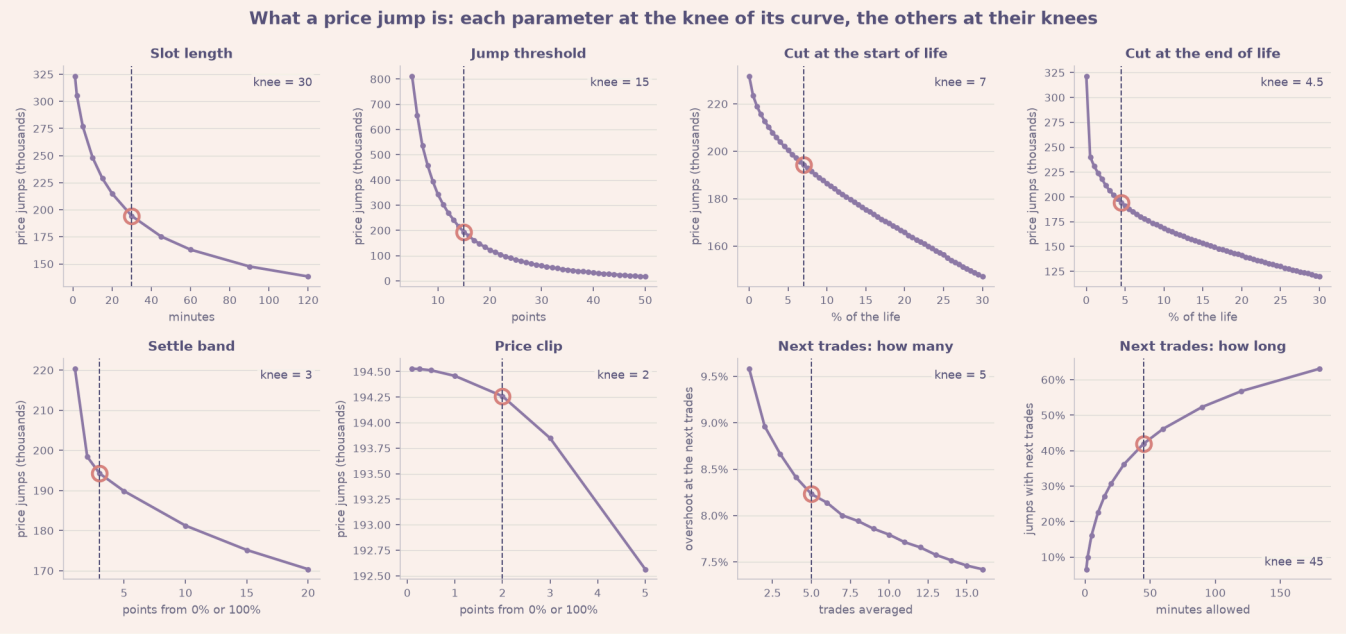
How the parameters were chosen: the elbow method

Every parameter of the analysis was set by the **elbow method**: how a price jump is defined, which part of a market's life is used, how the next trades are taken, the trader groups, the money threshold, the sample sizes, and the settings of each finding. Splits used only to display a finding (price bands, years, histogram bins) are not parameters and were not tuned; the uncertainty ranges are 95% ranges, a convention.

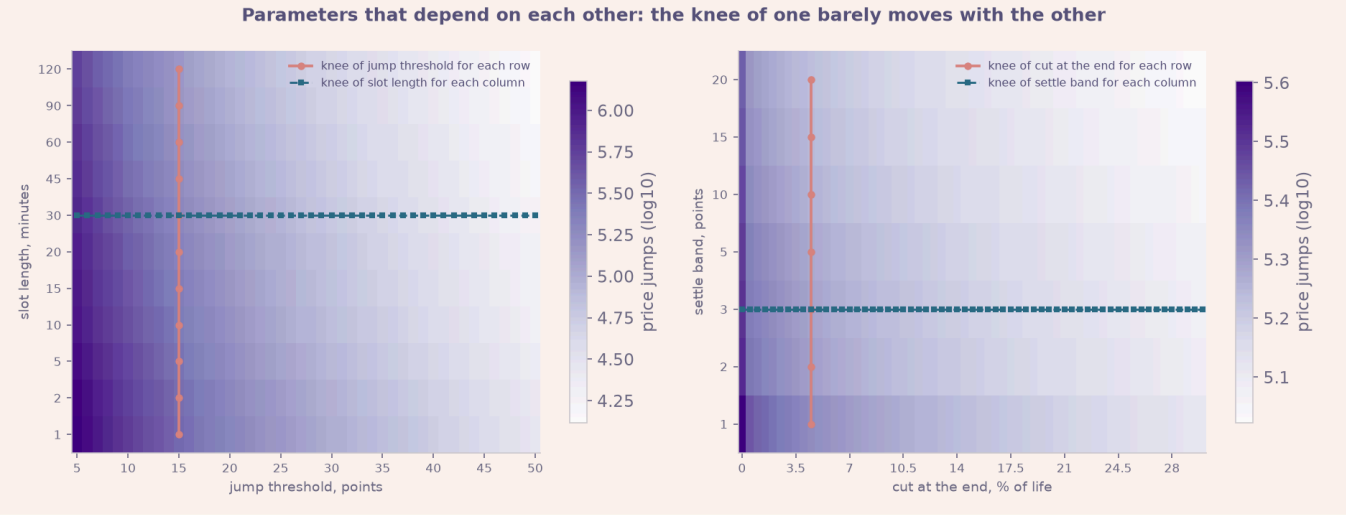
For each parameter, a curve is drawn over a wide range of values. It shows what the parameter controls: how many jumps it keeps, how noisy a number is, or the result itself. Such curves usually change fast at first and then flatten. The **knee** is where they bend; we find it with the Kneedle rule (Satopää et al., 2011): the point farthest from the straight line that joins the two ends of the curve, with both axes scaled to 0-1 (formula). Three rules apply to every curve:

- **One axis.** The knee is always found on a linear axis, never a log one (the two can give different knees for parameters tried over a wide range).
- **Enough jumps.** A point of a curve that is a share of jumps needs at least 1,000 jumps behind it; thinner points are left out.
- **Parameters that depend on each other are searched together.** Each one is set to the knee of its curve with all the others at their own knees, and this is repeated until no knee moves (a fixed point; two rounds for the event markets, three for the crypto ones).
- **Every market.** The curves are drawn on the whole dataset (the jump-definition search runs on the server, next to the data).

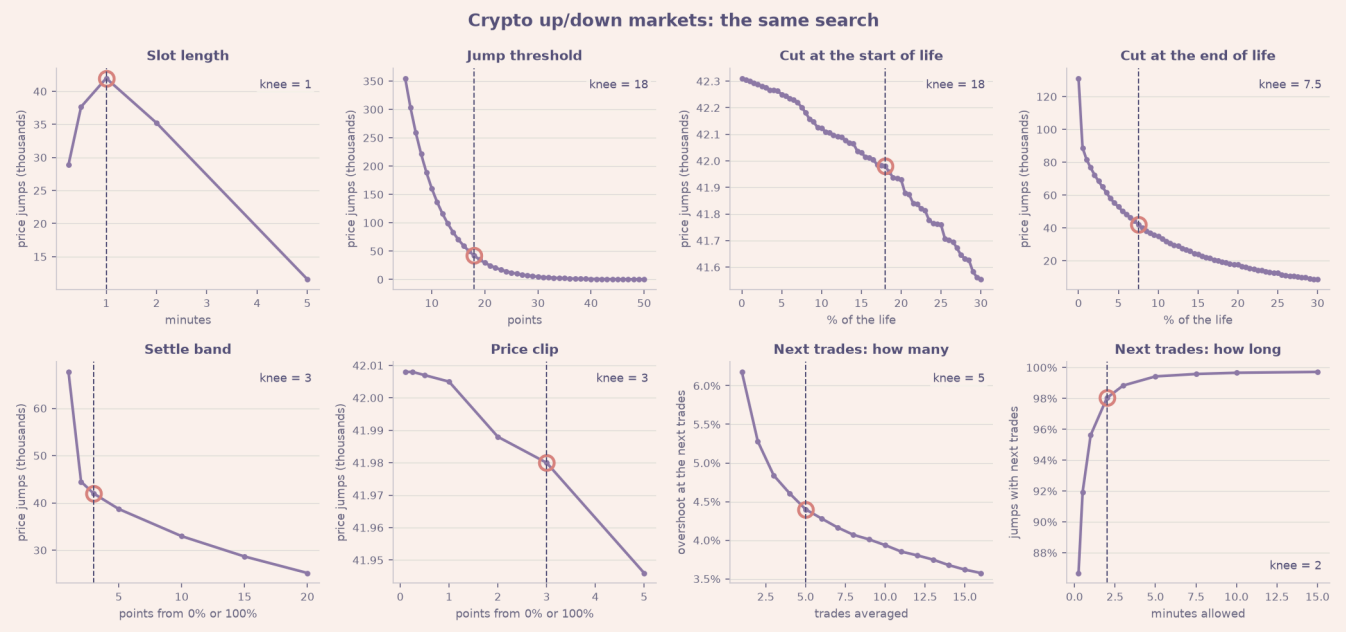
The parameters that define a price jump, at the fixed point (each curve drawn with the other parameters at their knees):

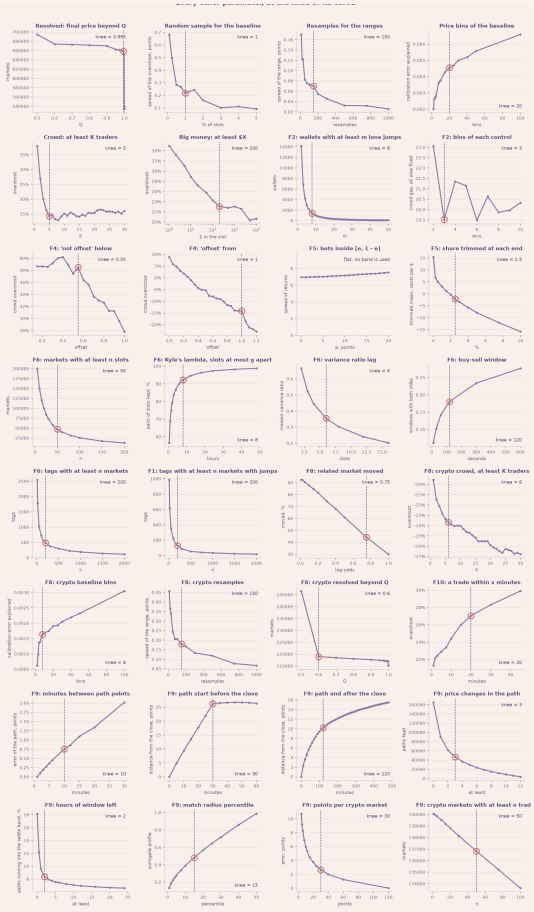


They do not depend on each other. For every slot length from 1 minute to 2 hours, the threshold's knee is 15 points; for every threshold, the slot length's knee is 30 minutes; the end cut's knee is 4.5% at every settle band, and the band's knee 3 points at every end cut:



The crypto up/down markets get their own search (F8), with their own trader groups, baseline bins, resamples and outcome rule; every other parameter has its own curve:





All the values are listed, with the range of each curve, in section 6.2. Three things to keep in mind:

- **Flat curves.** Where a parameter barely changes anything, its knee means little. The price clip changes the number of jumps by about 1% over its whole range, and the crypto start cut by under 2%. The price band of the betting test (F5) changes nothing at all, so no band is used. The pattern search (F9) uses every path that passes its filters, so it has no sample size.
- **Gentle bends.** Some curves are almost straight lines (the wait for a follow-up trade in F10, the "moved" threshold in F8, the match radius in F9). The number of resamples is a noise curve, which falls as $1/\sqrt{M}$: its knee is taken on that fitted shape. Their knee is a weak choice: another value close by would be as good.
- **Range.** A curve that falls off smoothly has no sharp bend, so its knee depends on the range of values tried. The ranges are stated in section 6.2.

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3. The findings

F1. Price jumps overshoot by over a third

Across 152,782 price jumps in 53,956 markets with a clear outcome, the price right after a jump is on average **too extreme by 36% of the jump** (range 35.1-35.9%).

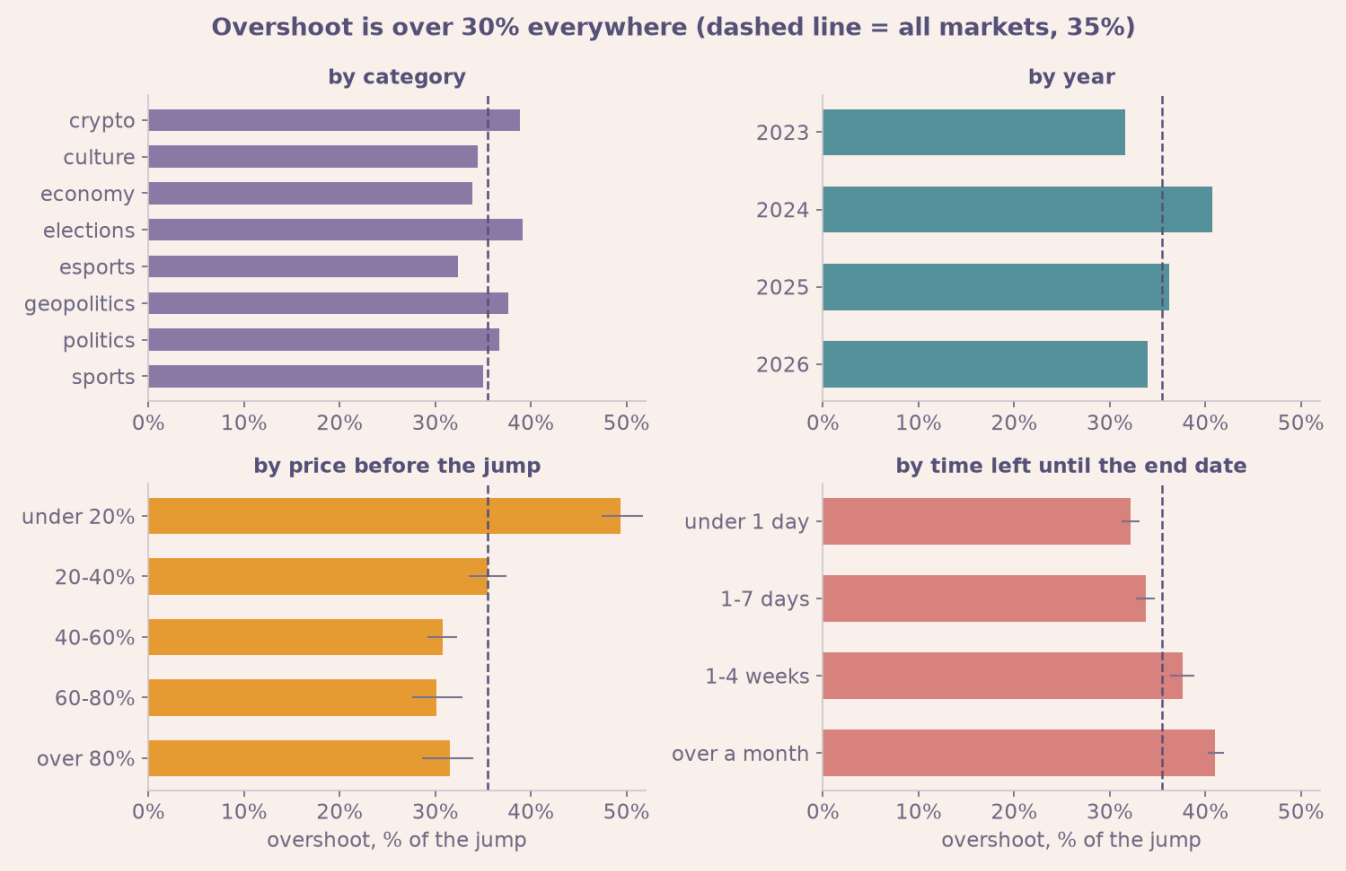
The plot shows it without any formula. For jumps that ended at a given price (x-axis), it shows how often YES actually won (y-axis):

- after an **up** jump to about 60%, YES won only 44% of the time;
- after a **down** jump to about 60%, YES won 63% of the time;
- at random moments, a 60% price won 58% of the time.

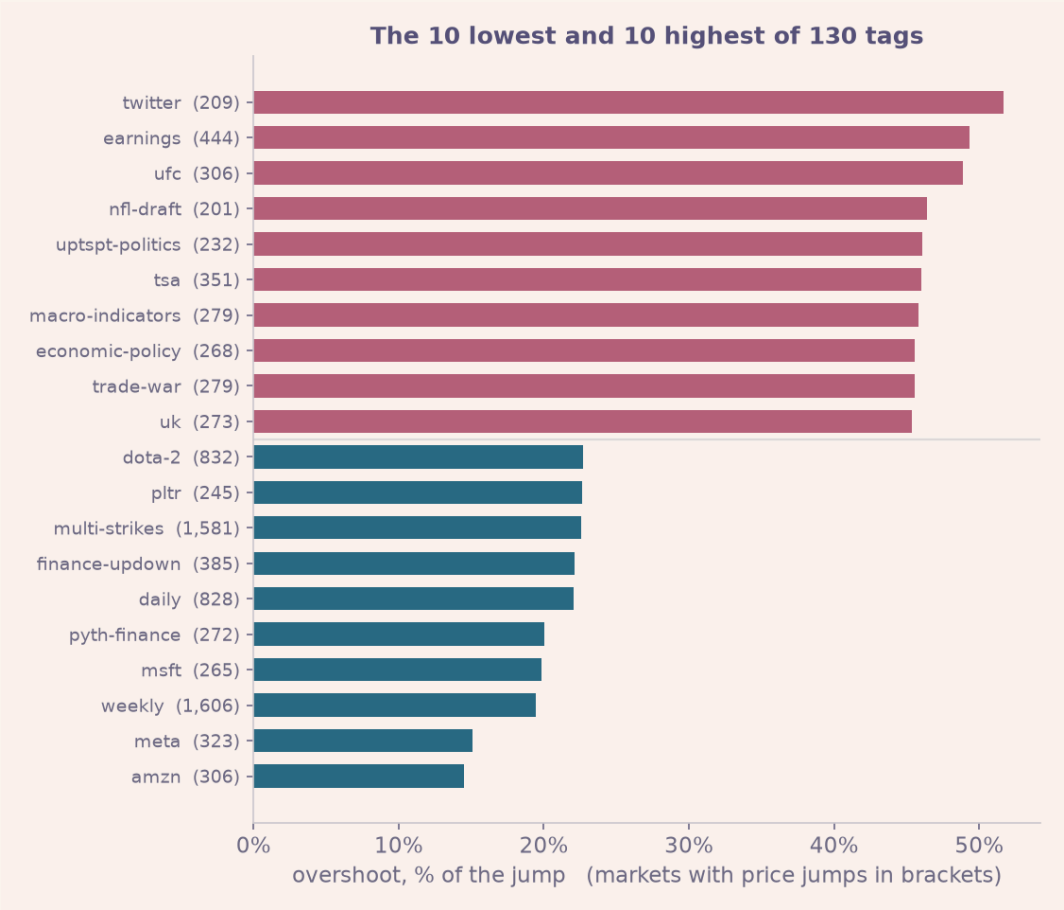
After a jump, the price sits too far in the direction of the jump.



It is between 32% and 41% in every category and every year, and at any distance from the market's end. It is highest for jumps starting below 20% (49%):

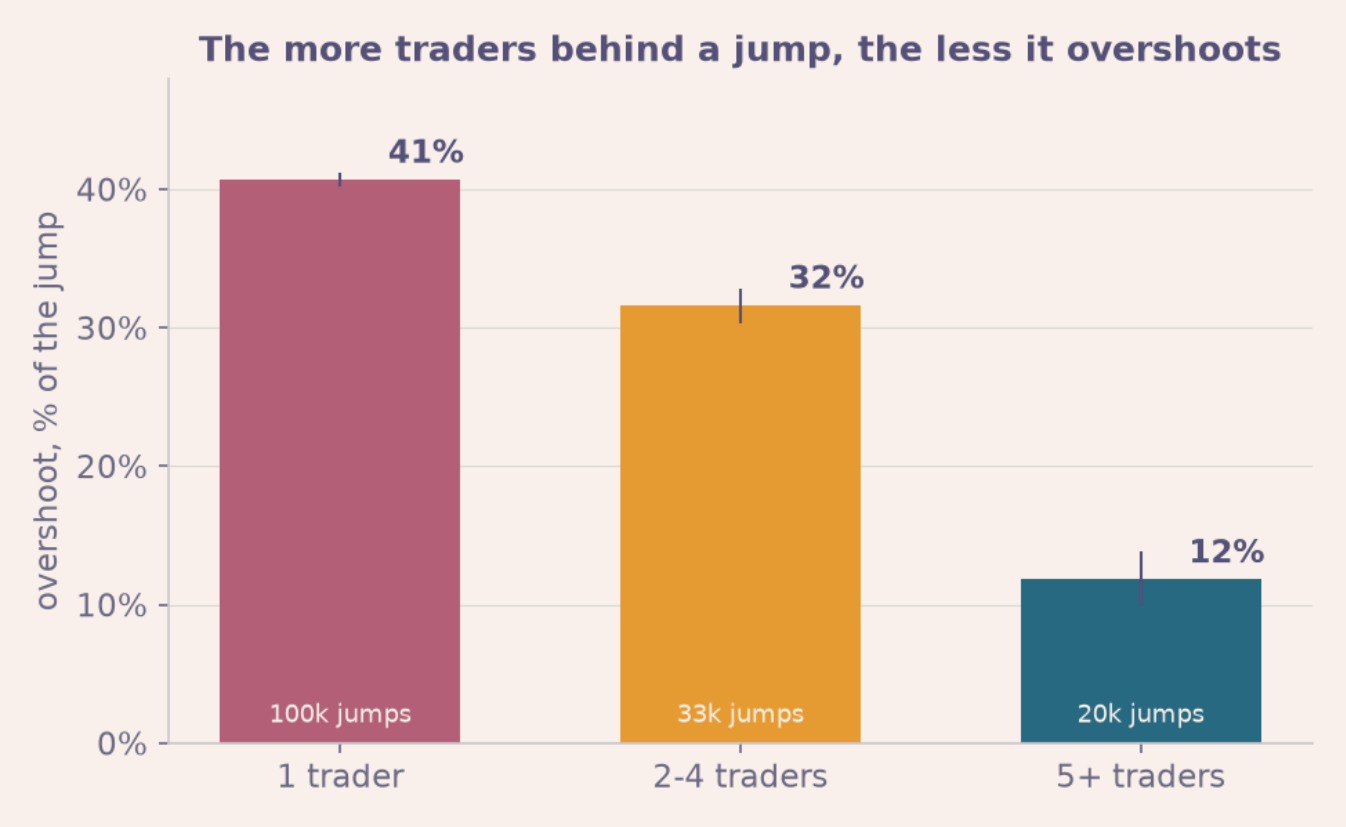


The size varies by topic. Across the 130 tags with at least 200 markets with jumps, it ranges from 15% to 52%. Markets on stock prices (Amazon, Meta, Microsoft) and weekly or daily price markets are at the low end; TSA, the NFL draft, UFC, earnings and Twitter at the high end.

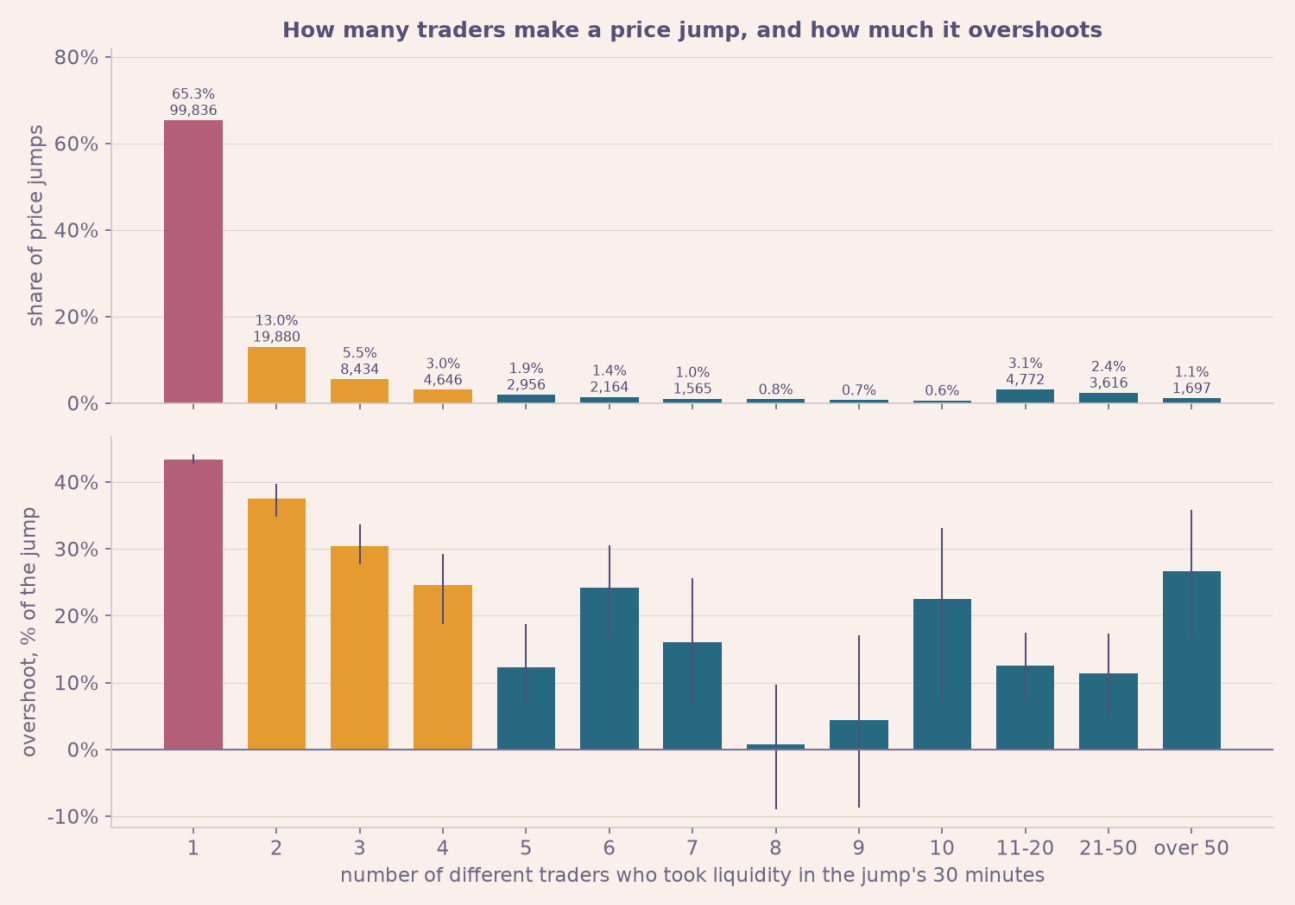


F2. Jumps made by many traders stick; jumps made by one trader don't

The best predictor of overshoot is **how many different traders took liquidity in the jump's 30 minutes**.

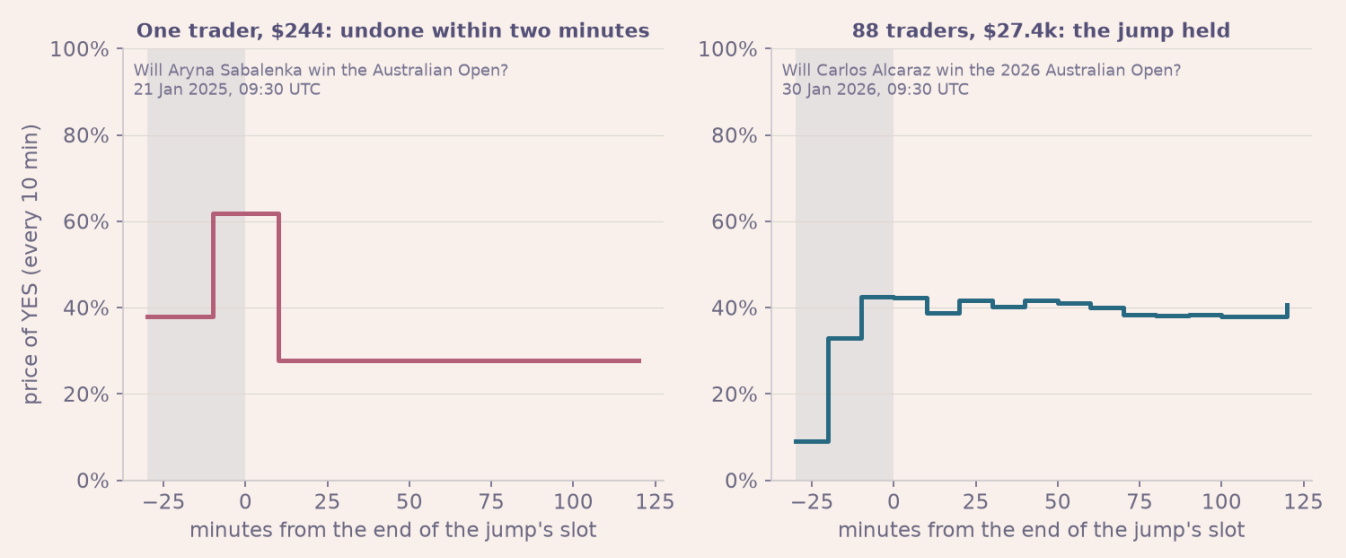


Trader by trader. 65% of price jumps are made by a single trader, 13% by two and 6% by three; the remaining 16% are spread over four traders and more, up to several hundred. The overshoot falls from 43% with one trader to 38% with two, 30% with three and 25% with four. The elbow of this curve is at five traders (section 2), so the report groups traders as 1, 2-4 and 5 or more (a "crowd"). From five on, each count has under 3,000 jumps and the values scatter (1% to 27%) with wide ranges; together they overshoot 14%.



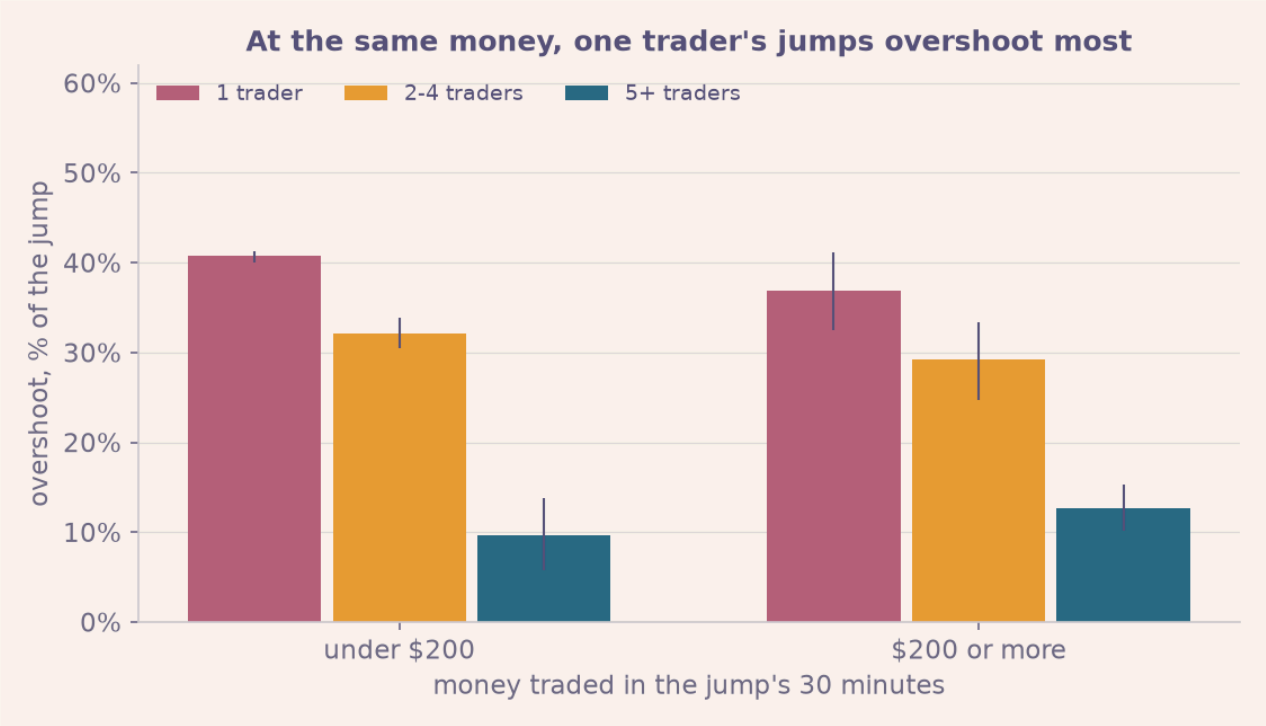
Two real jumps show the contrast:

- **Left:** on 21 January 2025, one trader paid $244 in 5 fills and pushed "Will Aryna Sabalenka win the Australian Open?" from 38% to 62%. The next trades, 74 seconds later, were at 29%, and the market resolved NO (she lost the final).
- **Right:** on 30 January 2026, 88 traders put $27,400 into "Will Carlos Alcaraz win the 2026 Australian Open?" and moved it from 14% to 35%. The price held and kept rising over the next hours, and the market resolved YES.

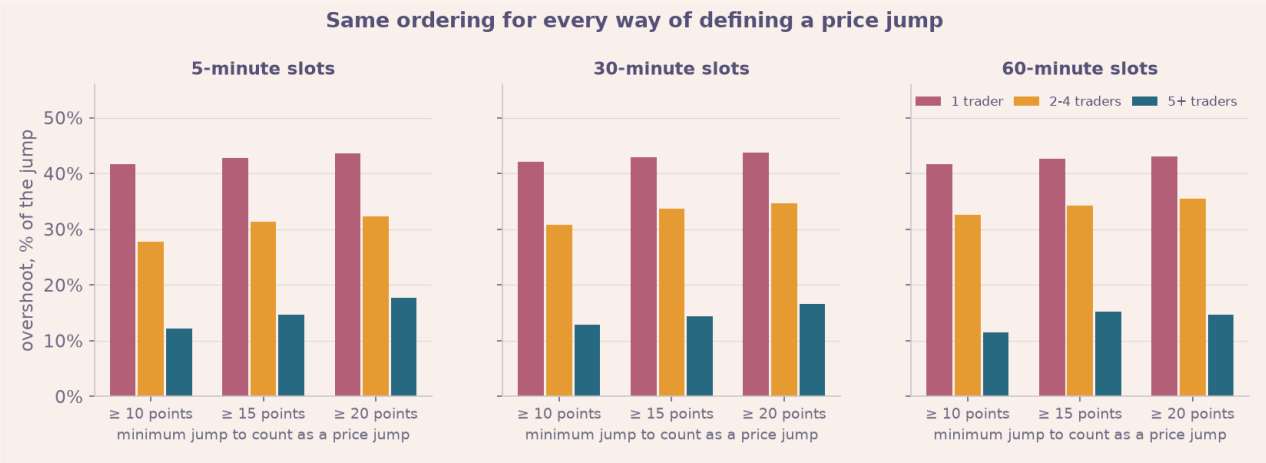


The pattern is robust:

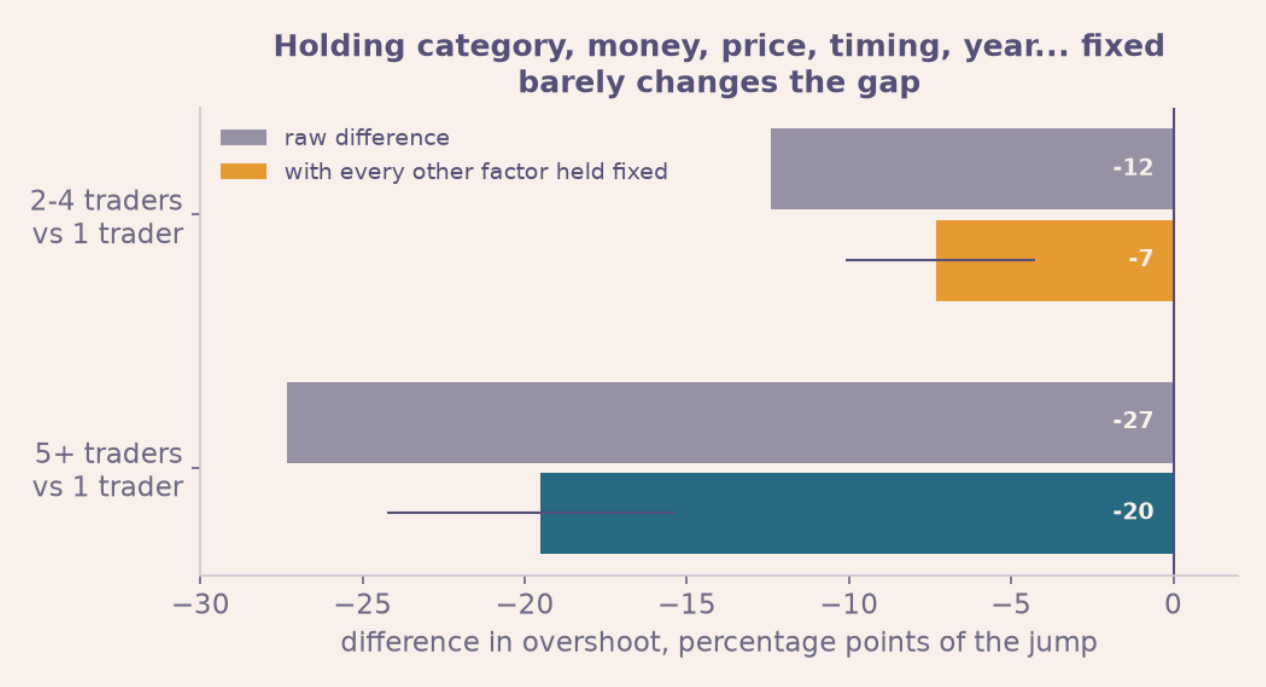
- **At the same money.** Below and above $200 traded in the slot (the money threshold set by the elbow method), one trader's jumps overshoot the most: 41% against 10% for a crowd below it, 37% against 13% above it.



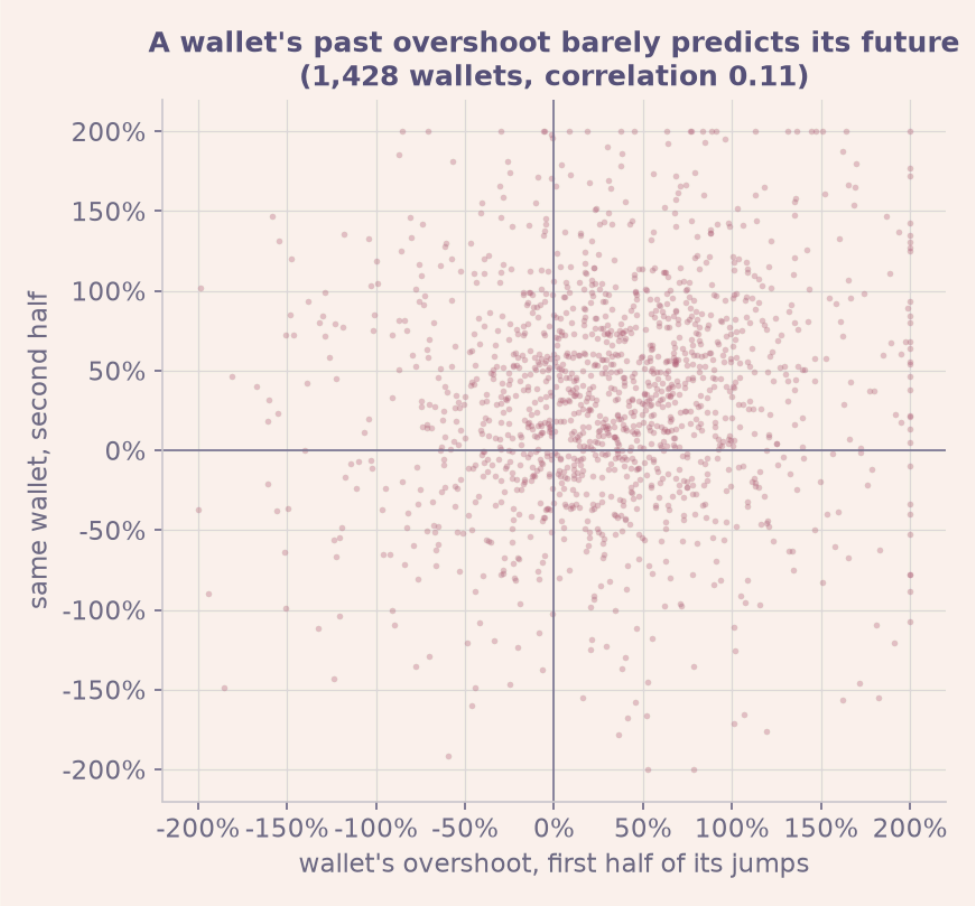
- **For other definitions of a price jump:** 5-, 30- or 60-minute slots, and jumps of at least 10, 15 or 20 points.



- **With everything else held fixed.** A regression holds category, question type, money, price level, time to the end, market lifetime, year and direction fixed at once. Crowd jumps still overshoot 20 points less than single-trader jumps, against 27 points raw.

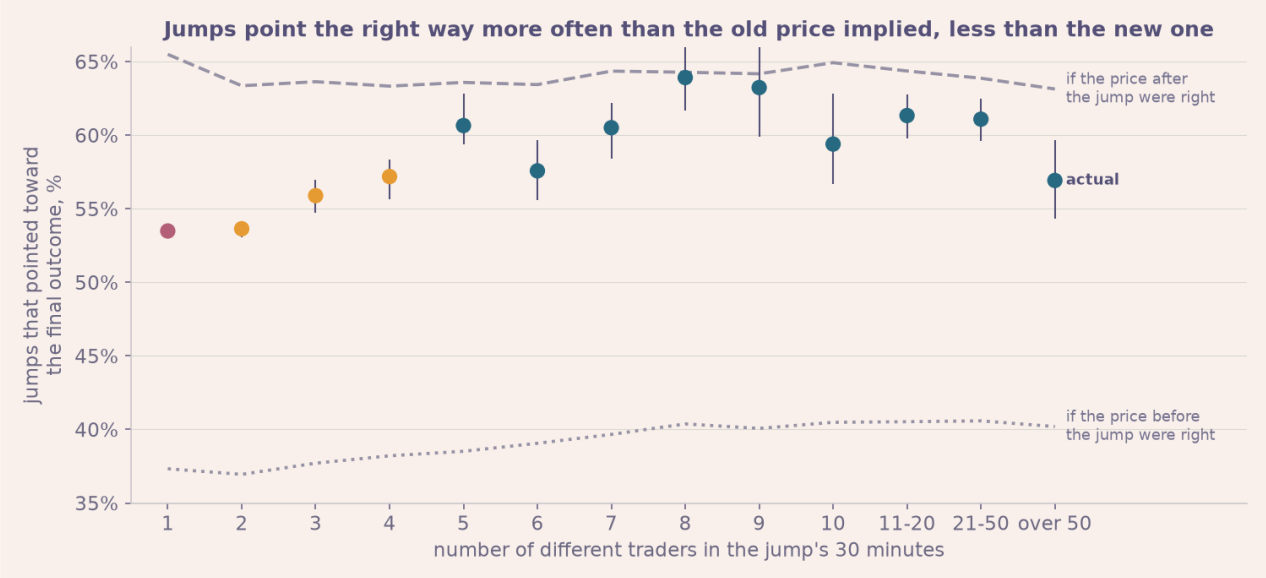


- **It's not a handful of wallets.** The 100,000 single-trader jumps come from 47,000 wallets. A wallet's overshoot in the first half of its history barely predicts the second half.



- **Do lone pushes point the right way?** A jump points the right way if it went up and YES won, or went down and NO won. Single-trader jumps did so 53.5% of the time. Had the price before the jump been right, they would have done so 37.3% of the time; had the price after it been right, 65.5%. So a lone push does carry information: it points the right way more often than the old price implied, but less

often than the new price claims. It covers (53.5 − 37.3) / (65.5 − 37.3) = 57% of the way, which is the 43% overshoot seen from another angle (why). Crowd jumps point the right way 60.5% of the time, against 39.9% and 64.0%: 85% of the way.

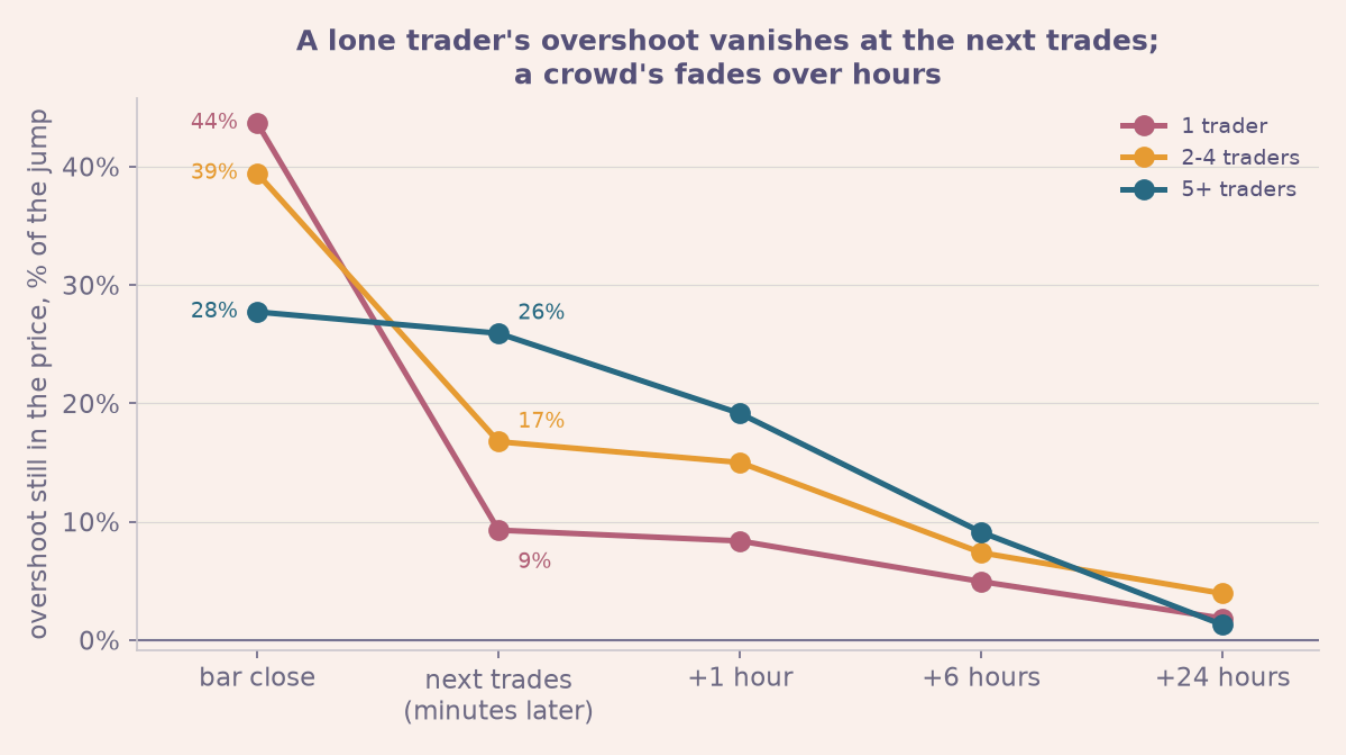


What it means. Agreement between several independent traders is a much better sign of information than the size of a jump.

F3. A lone trader's push is undone within minutes

Following the same jumps over time (those with prices a day later):

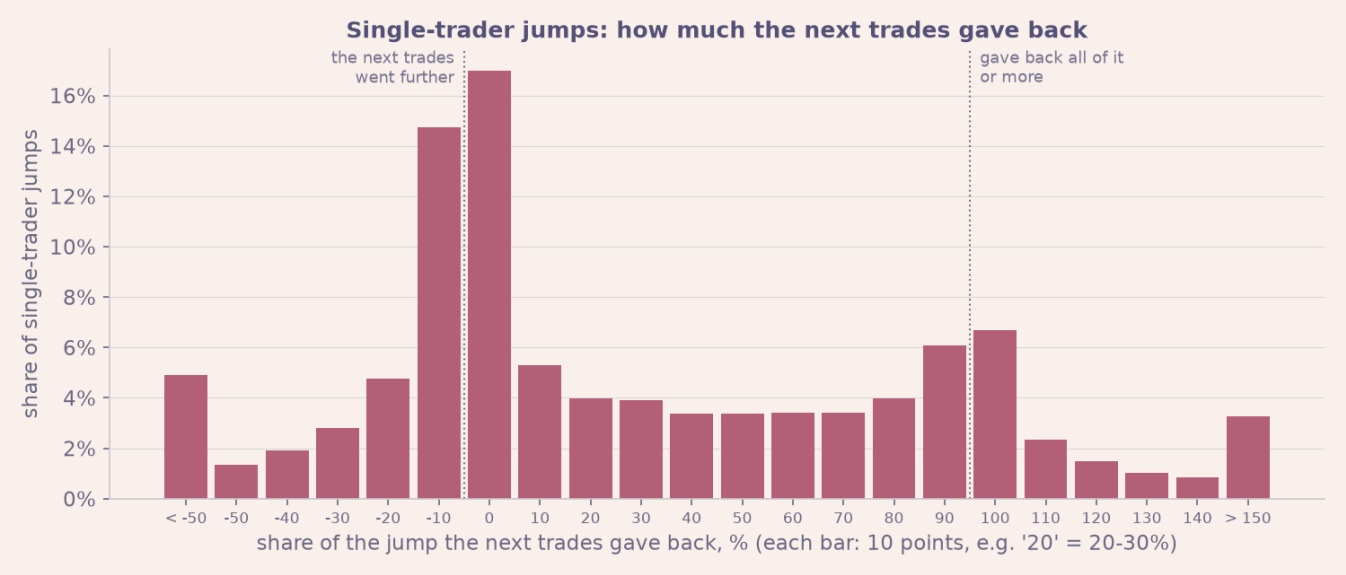
- **One trader:** the overshoot falls from 43% to **9% at the very next trades**.
- **A crowd:** it starts lower (28%) but lingers. It is still 26% at the next trades and 19% after an hour, and almost gone (1%) after a day.



"Next trades" means the first five trades within 45 minutes after the slot. They come a median of 2 minutes after crowd jumps and 15 minutes after single-trader jumps, which happen in quieter markets.

One jump at a time. The overshoot is an average: with an outcome that is either 0 or 1, it cannot be read off a single jump. What can be measured jump by jump is how much of it the next trades gave back. For the 28,190 single-trader jumps whose market traded again within 45 minutes (28% of them), the answer is spread out:

- 30% were taken further by the next trades;
- 17% were kept as they were (0-10% given back);
- 36% lost at least half, and 16% lost all of it or more;
- the median jump gave back 14%; weighted by size, 35% of the jump was given back.



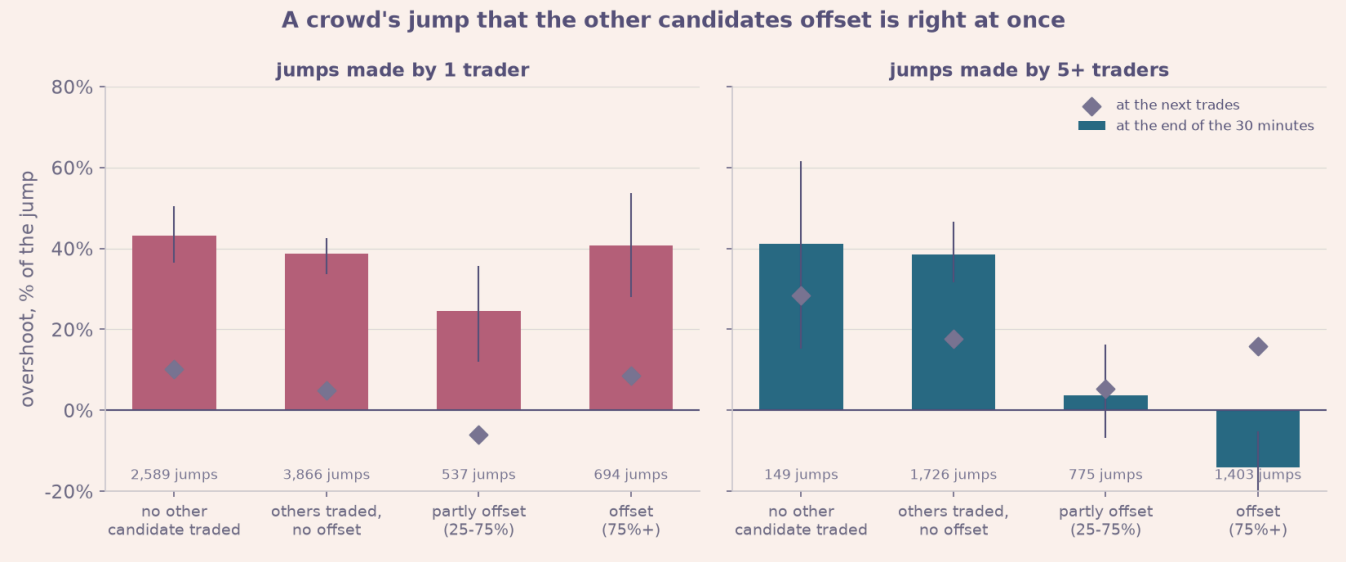
So the snap-back is not every lone push losing a bit: some are reversed completely, while others are kept or extended.

What it means. A lone trader's push is mostly temporary *price impact*: one order walks the price through a thin book, and the next traders bring it back. A crowd's smaller overshoot is a slower over-reaction.

F4. In a race, a crowd's jump that the other candidates offset does not overshoot

Many questions on Polymarket are races: who wins an election, which team wins a league. Each candidate is its own market, and exactly one of them will win. Most of these races are **neg-risk** events (a flag in the data): a NO share in one candidate can be converted into YES shares of all the others. This links the candidates' prices, so that their YES prices add up to about 100%, and it gives liquidity even to the long shots.

So when one candidate jumps, the others should fall by about as much in total. For each of the 17,106 price jumps in neg-risk events, we checked what the other candidates did in the same 30 minutes. They either **offset** the jump (moved the other way by at least as much as the jump), **partly** offset it (55-100%), did **not** offset it (less than 55%), or did not trade at all. The two cut-offs are the knees of the crowd's overshoot curve (section 2).



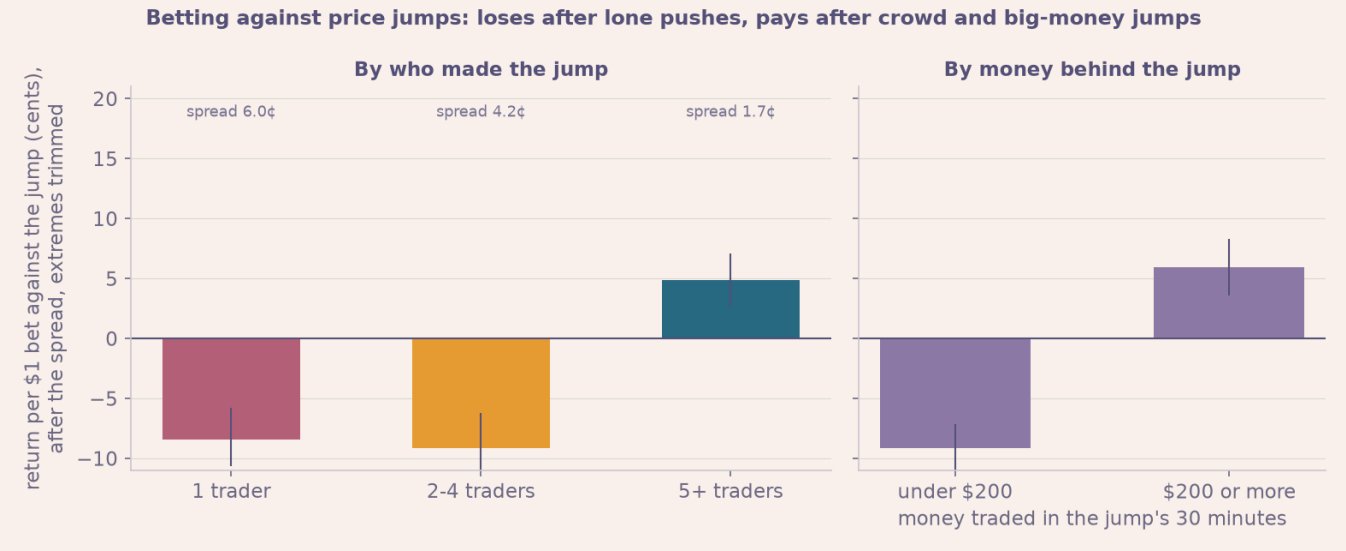
- A **crowd's jump** that the others offset does not go too far; if anything, it falls slightly short: −14% at the end of the 30 minutes (range −24% to −5%). Without the offset it overshoots by 39%, and by 41% if no other candidate traded.
- For a **lone trader's** jump the offset makes no clear difference (25-43%, with wide ranges).
- By the **next trades** the difference is gone: crowd jumps are 16-18% too far, offset or not.

What it means. In a race, a crowd jump that the rest of the field confirms by repricing is not too far. A jump the rest of the field ignores goes too far until the next trades correct it. Outside neg-risk events (37,000 jumps), offset jumps are no better (35% against 23% overshoot), but there the markets of an event need not exclude each other, so an "offset" means little.

F5. Betting against jumps: a loss after lone pushes, a small edge after crowd jumps

If jumps overshoot, betting against them should pay. Each test bet buys the opposite side right after the jump, at the price a bettor would actually have paid, and holds it to the end. Returns are measured per $1 staked, net of the usual pricing error.

A few very large wins dominate the plain averages, so the returns are trimmed: the most extreme 2.5% at each end are pulled in to the 2.5% limits (the knee of the trim curve). The level of the result depends on the trim, which keeps falling as more is trimmed; the ordering (crowd jumps far ahead of lone pushes) holds at every trim level.



- **After single-trader jumps, betting against them loses** about 9 cents per $1 (range 6-11). The fast snap-back happens, but the bid-ask spread (a median 6.0 cents) eats it.
- **After crowd jumps, it gains** about 5 cents per $1 (range 3-7), where the spread is only 1.7 cents.
- **After jumps with $200 or more behind them, it gains** about 6 cents per $1 (range 4-8).

What it means. The large overshoot of lone pushes is gone before anyone can trade on it. What a bettor can capture is the smaller overshoot of crowd and big-money jumps, which the next trades leave in place (F3). Fees and queue position are not modelled, so this is an upper bound.

F6. Lone pushes happen in thin markets

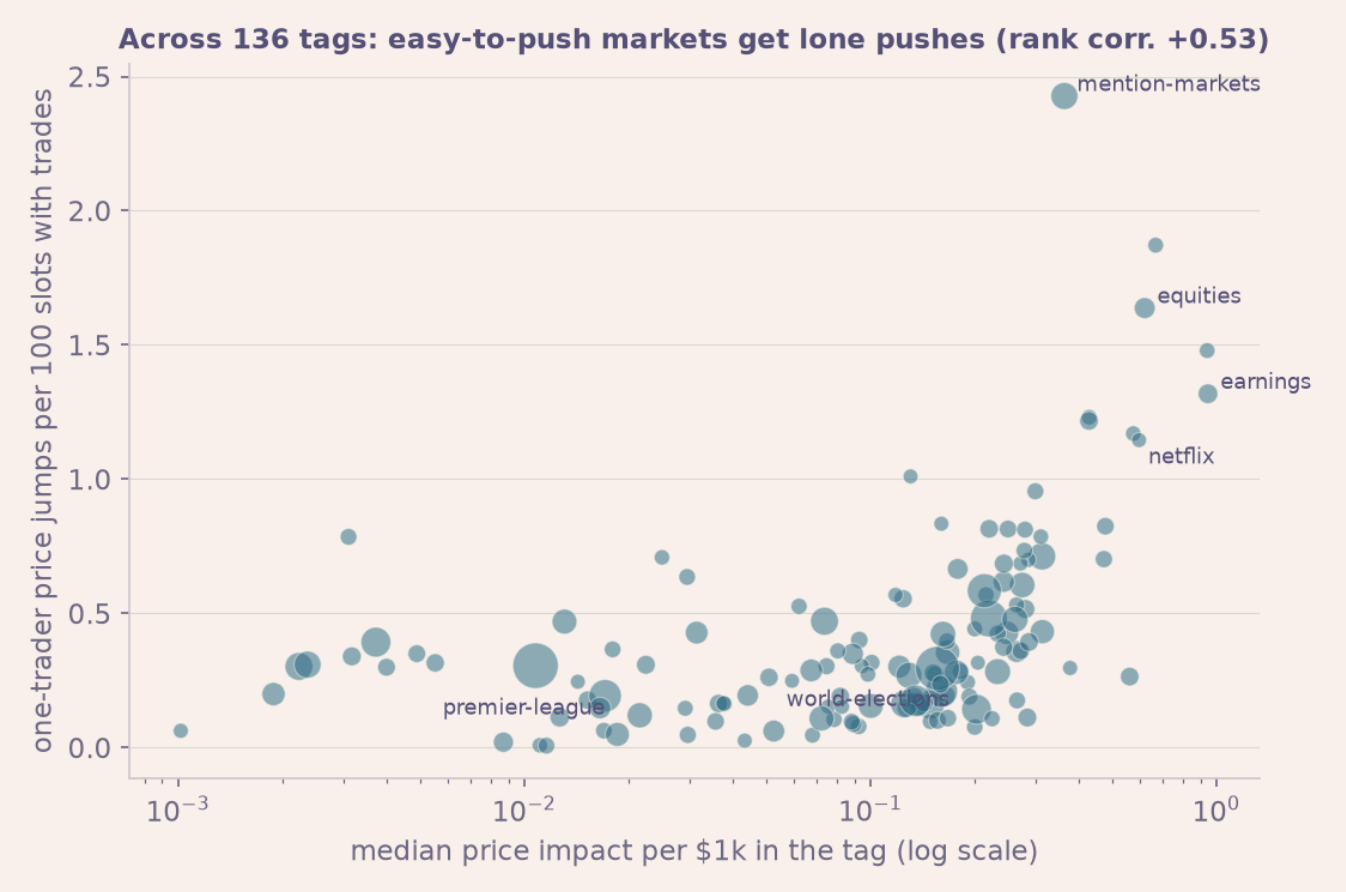
Some markets see single-trader jumps all the time, others never. For each of 47,118 markets with at least 50 slots with trades, standard market-structure measures were computed:

- how concentrated trading is among wallets (Gini, Herfindahl, the biggest wallet's share);
- how far $1,000 of net buying moves the price (Kyle's lambda);
- the bid-ask spread (Roll's estimator, and the buy-sell price gap within a minute);
- how balanced buying and selling are.



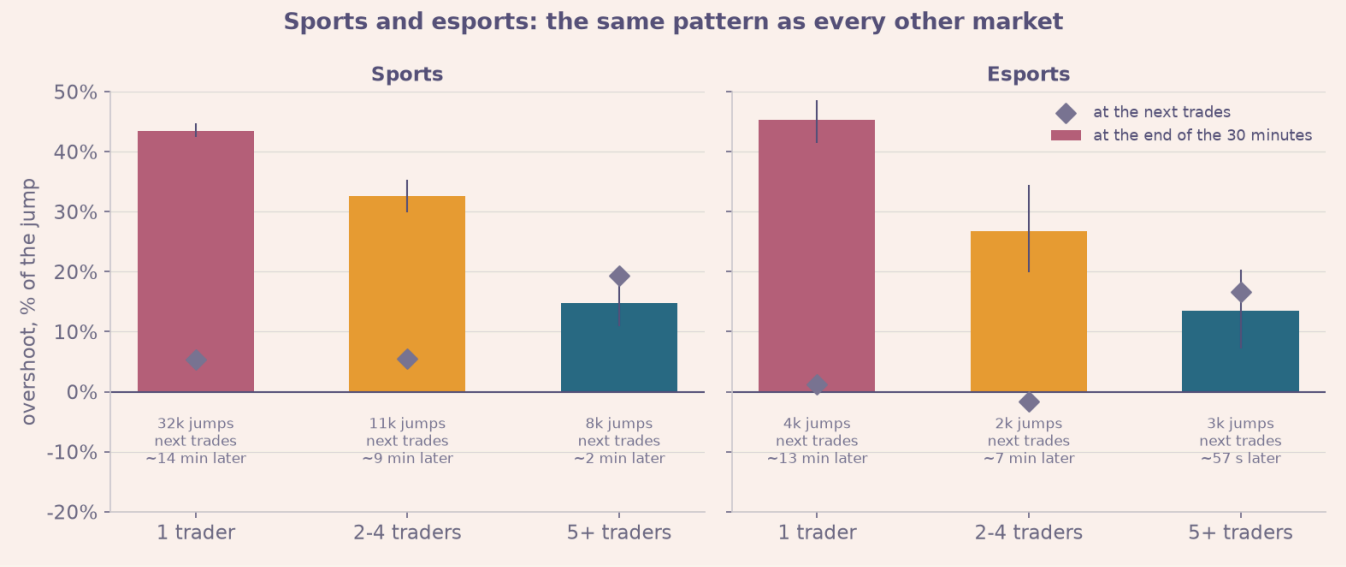
- **How often one trader can push the price depends on how thin the book is.** With market size held fixed, it correlates with the spread (+0.54) and with the price impact per dollar (+0.17).
- **Concentration among a few wallets doesn't explain it** (Gini −0.13: if anything, the opposite).
- **This holds in every category.**
- **How far the push goes doesn't depend on the market:** one-trader jumps overshoot 42-51% in every fifth of the markets, on every measure (right panel).

Across the 136 tags with at least 200 markets the link is strong (+0.53). Mention markets, Apple, equity and stock markets, approval ratings and earnings are at the top; European football and elections at the bottom.



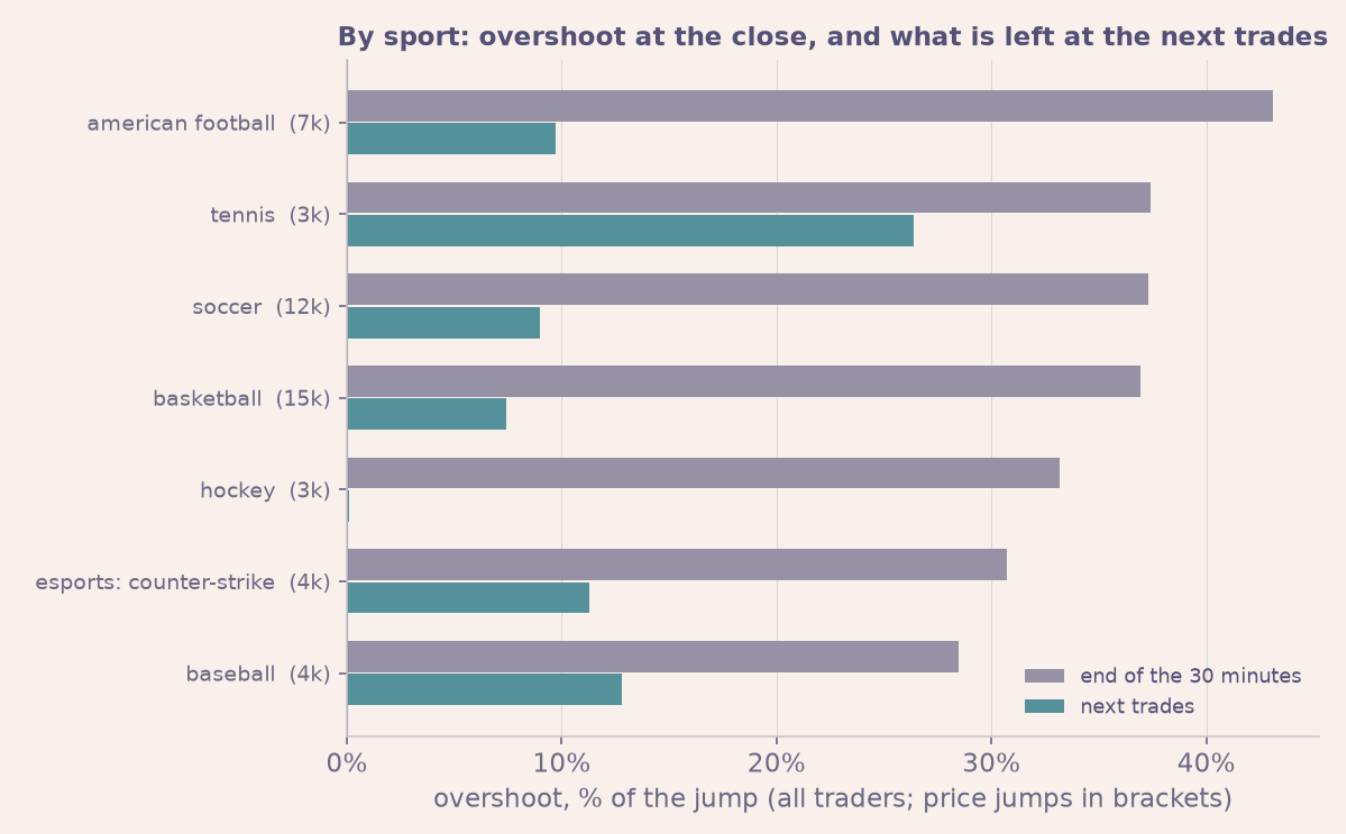
F7. Sports and esports: the same pattern

Sports and esports make up most markets but have the fewest jumps per market: 59,128 price jumps in 29,146 markets with a clear outcome. Because the left-out stretches scale with each market's life, a game market keeps its game. The data has no kickoff times, so jumps before and during a game are not told apart.



- **Lone pushes overshoot as everywhere else** (43% in sports, 45% in esports), and the next trades undo them.
- **Crowd jumps overshoot less** (15% and 14%), but 17-19% is still there at the next trades, which come within a minute or two.

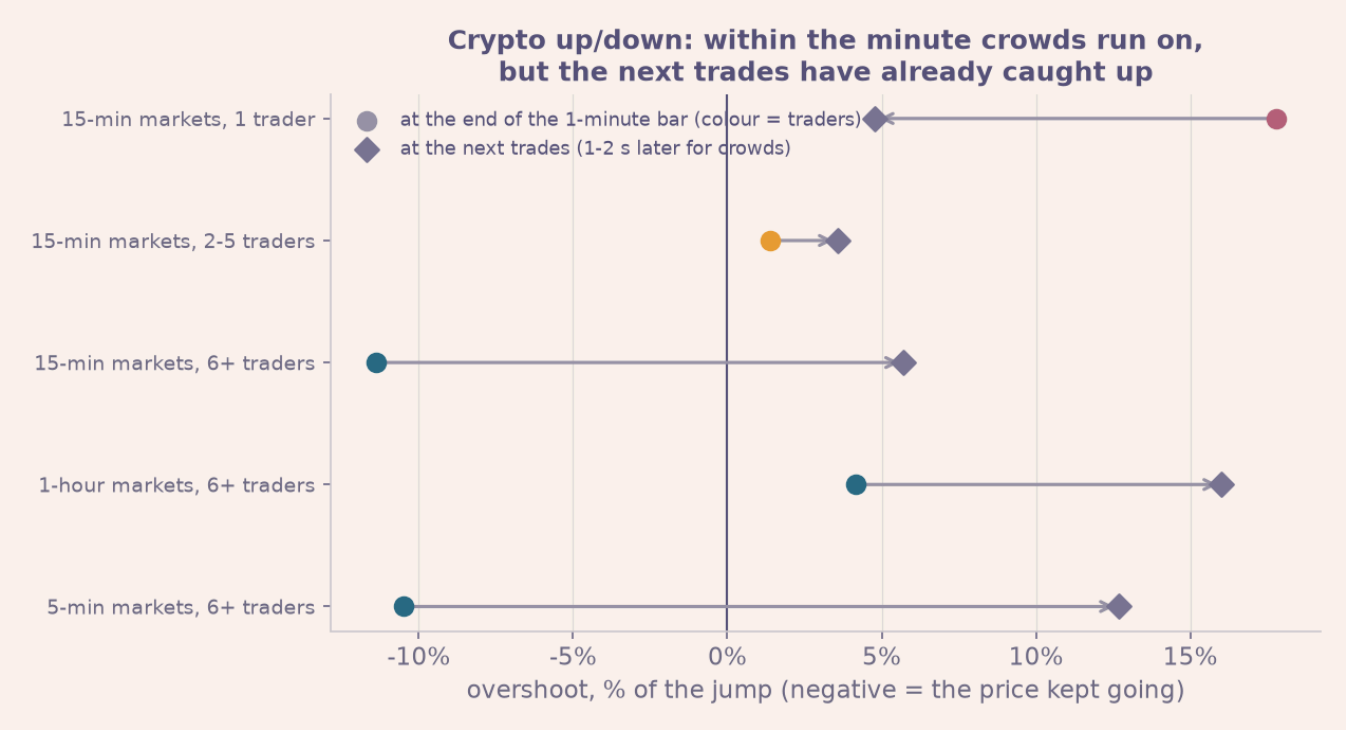
By sport, the overshoot ranges from 22% (Dota) to 51% (combat sports):



What it means. Game markets behave like every other market.

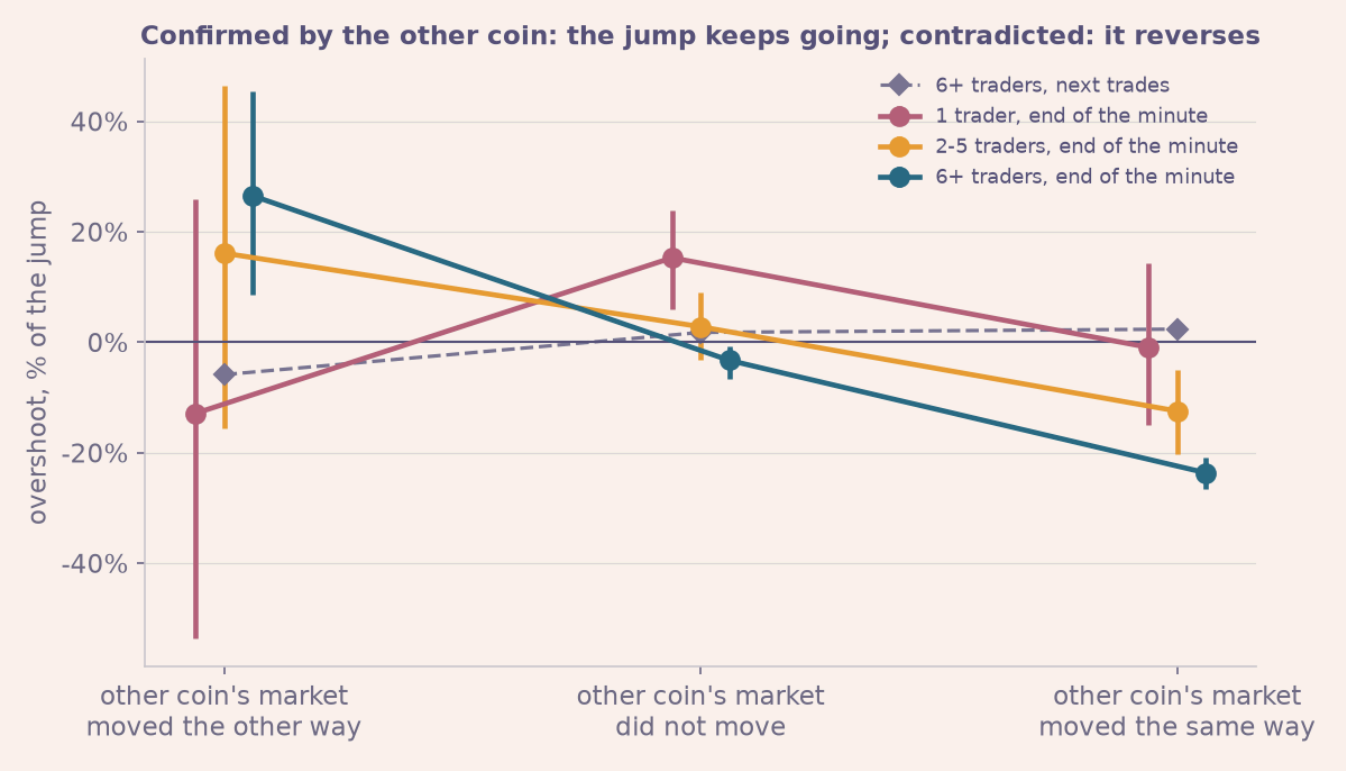
F8. Crypto "up or down" markets: the same pattern, over within seconds

These recurring markets (bitcoin, ether, solana and xrp, over 5 minutes, 15 minutes or 1 hour) hold 59% of all trades. They get their own elbow search, which gives 1-minute slots, jumps of at least 18 points, the first 18% and the last 7.5% of each market's life left out, a 3-point settle band and price clip, next trades of 5 trades within 2 minutes, trader groups of 1, 2-5 and 6 or more (a crypto crowd), 8 baseline price bins, and an outcome rule of 0.6 (these markets settle at exactly 0, 1 or a split). That leaves 36,161 price jumps in 27,663 markets.



At the end of the minute the familiar pattern appears: lone pushes go too far (+18% in the 15-minute markets) and crowd jumps keep going (−10% to −11% in the 5- and 15-minute markets). By the next trades, seconds later, the ordering is gone: every group sits between +4% and +16%.

A related market decides the direction. A crowd jump keeps going when the other coin's market of the same length moved the same way in that minute (−24%), and reverses when it moved the other way (+27%).



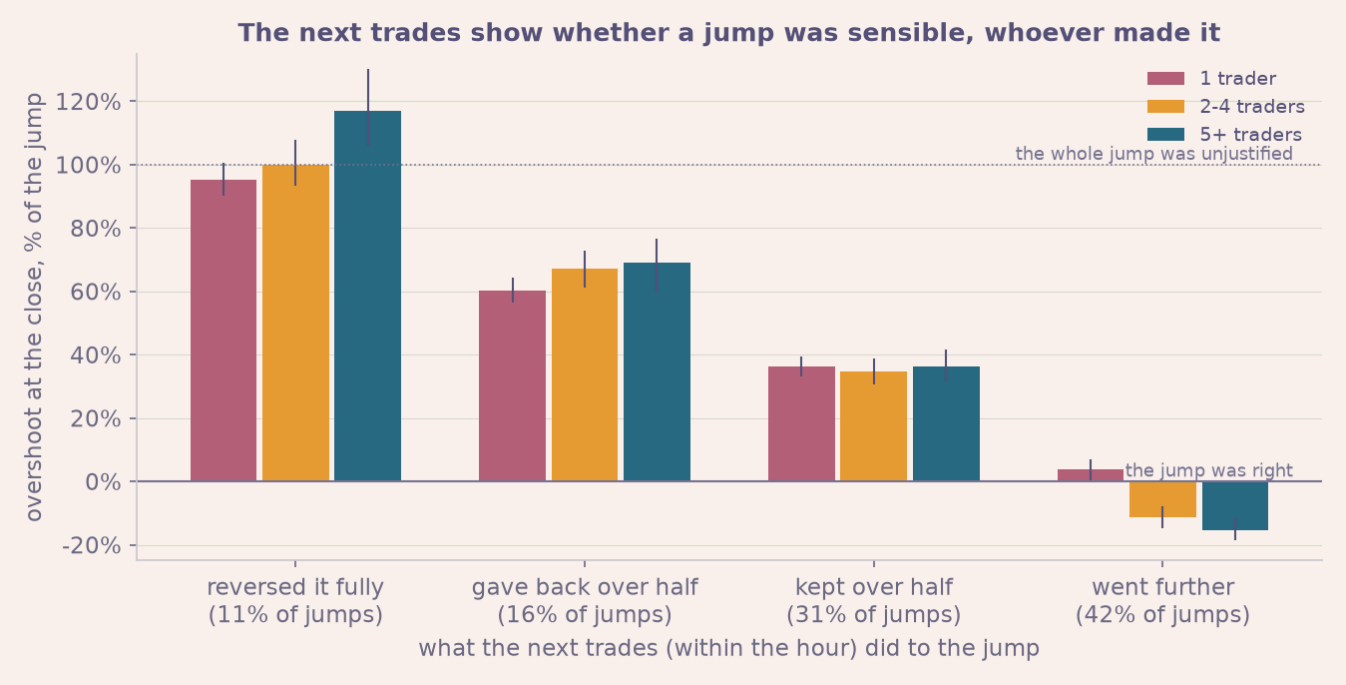
F9. Searching for recurring patterns

We also looked for recurring shapes with *matrix profiles*, a standard pattern-search tool for time series. Price paths after price jumps show **almost no recurring shape beyond chance**: 15.4% of real paths have a close match elsewhere, against 15.0% of shuffled ones. In the chain of bitcoin's 15-minute markets, the unusual stretches are "whipsaw" markets where the price crosses 50% about six times. Comparing the coins' chains shows that their markets move together, with bitcoin and ether about 30 seconds ahead of solana and xrp. All of this is real but has plain explanations, and none of it is tradable.

F10. News or noise? A tentative look

A last, less conclusive check: can one tell a jump that carries news (many traders, a lasting effect) from one trader's noise?

After the fact, partly. What the next trades within 45 minutes do to a jump says more than who made it. Jumps the next trades took further were about right at the close; jumps they reversed were wholly unjustified. This holds for every trader group. A crowd jump that the next trades kept had no overshoot at all (0%, range −2% to 3%). A one-trader jump that the next trades mostly gave back had 75%.



At the moment of the jump, though, the signs are weak. The number of traders helps most (14% overshoot with a crowd, 41% without). $200 or more helps too (23% vs 41%), and so does a new trade within 20 minutes (27% vs 43%). Another market of the same event jumping at the same time does not help (35% vs 39%). And 72% of one-trader jumps have no trade at all in the following 45 minutes, so for most lone pushes the after-the-fact test cannot be applied. This is a description, not a tool.

4. What did not work

- **Recurring price shapes across markets.** Real matches were barely more common than chance.
- **"Price moved without money" anomalies.** Mostly bid-ask bounce: trades alternating between the buy and the sell side.
- **"More trades means more reversal".** An artefact of using the last trade price: a sweep's last fill is its most extreme price. Fixed by using slot averages.
- **"Bet against crypto YES spikes".** It was just crypto YES being about 8 points overpriced in general. Fixed by measuring the usual error per category.
- **Concentration (Gini) as the cause of lone pushes.** With size and price impact held fixed, it adds nothing.
- **Bitcoin anomalies clustering at US data-release hours.** This did not hold up.
- **Splitting game markets into before and during the game.** The data has no kickoff times, and the end date is not a reliable stand-in (esports matches often settle hours before it), so no such split is made.

5. Limitations

- **"Next trades" is an average of real trades, not a guaranteed price.** F5 uses side-specific prices, but fees and queue position are not modelled.
- **The usual-error correction is estimated from the data**, per category or question type. Other ways of estimating it move the levels by 1-3 points without changing any comparison.
- **Most jumps are small in money.** The findings describe how prices behave, not a large pool of money.
- **The last 4.5% of a long market is left out.** For an election market that is its final weeks, when much of the news arrives.
- **Approximate labels.** Categories and question types come from text rules.
- **No external data.** There are no news timestamps or live scores, so "confirmation" always means other Polymarket prices.

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6. Reference: formulas, methods and parameters

6.1 Formulas

Every quantity in the report is defined here. Each entry says what the formula is for, gives it, explains every symbol, and says how to read the result. The values of the parameters (Greek letters and capitals such as Δ, θ, N) are in section 6.2, with the elbow curve that set each one.

Notation. A trade i has a time t_i (seconds), a YES price y_i (between 0 and 1), a dollar amount u_i and the wallet w_i of its taker (the trader who accepted a posted order). A market's final outcome is $O = 1$ if YES won and $O = 0$ if NO won. $\mathbb{1}[\cdot]$ is 1 when the condition in brackets is true and 0 otherwise; $|S|$ is the number of elements of a set S.

Prices and price jumps

YES price of a trade. A market has two tokens, YES and NO, whose prices add up to about 1. To put every trade on one scale, a trade in the NO token is turned into the YES price:

$$y_i = \begin{cases} \text{price}_i & \text{if trade } i \text{ is in the YES token} \\ 1 - \text{price}_i & \text{if it is in the NO token} \end{cases}$$

Slots. Time is cut into slots of length Δ. Trade i belongs to slot $s = \lfloor t_i/\Delta \rfloor$ (the whole part of the division), and the slot is stamped with its closing time $t_s = (s+1)\Delta$. Only slots with at least one trade are used.

Slot price, money and traders. The slot's price is the volume-weighted average price (VWAP) of its trades: each trade counts in proportion to its dollars, so a $100 trade weighs a hundred times more than a $1 trade. The slot's money is the dollars traded, and its number of traders is the number of different taker wallets:

$$p_s = \frac{\sum_{i \in s} u_i y_i}{\sum_{i \in s} u_i}, \qquad U_s = \sum_{i \in s} u_i, \qquad K_s = \left|\{w_i : i \in s\}\right|$$

Price clip. Prices very close to 0 or 1 are pulled in to $[\kappa, 1-\kappa]$, so that a move from 0.2% to 0.1% is not treated as meaningful and log-odds (below) stay finite:

$$\tilde{p} = \min\left(\max(p, \kappa),\ 1 - \kappa\right)$$

Price jump. Let s^- be the previous slot of the same market that had trades. The change and the jump rule are

$$\Delta p_s = \tilde{p}_s - p_{s^-}, \qquad \text{slot } s \text{ is a price jump if } |\Delta p_s| \geq \theta$$

θ is in price points (0.16 = 16 percentage points). The jump's direction is $d_s = \text{sign}(\Delta p_s)$: +1 for up, −1 for down. For a jump j we write p_{j^-} for the price before, p_j for the price after (the slot's clipped price) and $|\Delta p_j|$ for its size.

Log-odds. Some steps work on the log-odds scale, which stretches prices near 0 and 1:

$$\ell(p) = \log\frac{\tilde{p}}{1 - \tilde{p}}$$

A price of 50% has log-odds 0; 90% has 2.2; 99% has 4.6.

Which part of a market's life is used

Settle moment. A market's price eventually runs to 0 or 1 as its outcome becomes known. Its settle moment t_{end} is the close of its last slot whose last trade was inside $[\varepsilon, 1-\varepsilon]$, if the market's final slot is outside that band; otherwise it is the close of its last slot. Slots after t_{end} are not used.

Position in the life. With t_{first} the close of the market's first slot, the position of slot s in the market's life is

$$f_s = \frac{t_s - t_{\text{first}}}{t_{\text{end}} - t_{\text{first}}}$$

so $f = 0$ is the first slot and $f = 1$ the settle moment. A slot is used only if

$$c_{\text{start}} \le f_s \le 1 - c_{\text{end}}$$

which leaves out the first c_{start} and the last c_{end} of every market's life, whatever its length.

Outcome. From the market's final outcome price q_m (the first of its two outcome prices): $O = 1$ if $q_m > Q$, $O = 0$ if $q_m < 1 - Q$; markets in between (cancelled or split) are left out.

Overshoot

Usual error (baseline). Prices are slightly off even at quiet moments. To measure this, a share ρ of all used slots is drawn at random. For each category c (politics, sports, …) and each of W equal price bins B, the usual error is the average gap between outcome and price:

$$b_c(B) = \frac{1}{|R_{c,B}|} \sum_{r \in R_{c,B}} \left(O_r - p_r \right)$$

where $R_{c,B}$ is the set of random slots of category c whose price is in bin B. A negative value means prices in that bin are usually too high. For a price p, $b_c(p)$ is the value of the bin that contains p.

Overshoot of one jump. Measured at a later price p (the price right after the jump, the next trades, or the price h hours later), in price points:

$$o_j(p) = -\left[(O_j - p) - b_c(p) \right] \cdot d_j$$

$O_j - p$ is how far the outcome landed from the price; subtracting $b_c(p)$ removes the usual error; multiplying by $-d_j$ makes the result positive when the price was too far *in the direction of the jump*. Example: an up jump to 60% in a market that resolved NO, with a usual error of −2 points, gives
$o = -[(0 - 0.60) - (-0.02)] \cdot 1 = 0.58$.

Overshoot share. The number the report calls "overshoot", for a group J of jumps:

$$\text{OS}(J) = \frac{\sum_{j \in J} o_j(p)}{\sum_{j \in J} |\Delta p_j|}$$

It is the part of the total jump size that the outcomes did not justify. 0% means the jumps were right on average; 100% means they were wholly unjustified; below 0% means the price did not go far enough. One jump's $o_j / |\Delta p_j|$ means little on its own: with an outcome of 0 or 1 it is either far above or far below zero. Only the share over many jumps is informative.

After the jump

Next trades. The first N trades of the market after the slot closes, within T seconds, averaged by dollars:

$$p_j^{\text{next}} = \frac{\sum_{k=1}^{N} u_k y_k}{\sum_{k=1}^{N} u_k}$$

This is the first price an outsider could have traded at after seeing the jump. Jumps with no trade within T have no next-trades price.

How the price settles (F3). The share of the jump still in the price h hours later, on the log-odds scale, using the last slot with trades before $t_j + h$:

$$R_j(h) = \frac{\ell(p_{t_j+h}) - \ell(p_{j^-})}{\ell(p_j) - \ell(p_{j^-})}$$

$R = 1$: the whole jump is still there; $R = 0$: the price is back where it started. The decay curve in F3 is OS computed at the close, at p^{next}, and at p_{t_j+h} for h = 1, 6 and 24 hours, on the same jumps (those with all horizons inside the window).

Share of a jump given back by the next trades:

$$g_j = \frac{p_j - p_j^{\text{next}}}{p_j - p_{j^-}}$$

$g = 0$: the next trades kept the jump; $g = 1$: they undid all of it; $g < 0$: they went further. The F10 reactions use $1 - g_j$, the share kept: below 0 "reversed it fully", 0-0.5 "gave back over half", 0.5-1 "kept over half", above 1 "went further".

Pointing the right way

Accuracy of a group of jumps. A jump points toward the outcome if it went up and YES won, or went down and NO won. The share of a group J that did so:

$$A(J) = \frac{1}{|J|} \sum_{j \in J} \mathbb{1}\left[d_j = 2O_j - 1\right]$$

($2O_j - 1$ is +1 when YES won and −1 when NO won.)

What a price implies. If prices were exactly right, an up jump to p_j would point the right way with probability p_j (the chance of YES), and a down jump with probability $1 - p_j$. The rate implied by the price after the jump, and the same with the price before it:

$$E_{\text{after}}(J) = \frac{1}{|J|} \sum_{j \in J} \begin{cases} p_j & d_j = +1 \\ 1 - p_j & d_j = -1 \end{cases}, \qquad E_{\text{before}}(J) = \frac{1}{|J|} \sum_{j \in J} \begin{cases} p_{j^-} & d_j = +1 \\ 1 - p_{j^-} & d_j = -1 \end{cases}$$

Share of the way. How far the jumps went toward being right, from the old price ($S = 0$) to the new one ($S = 1$):

$$S(J) = \frac{A(J) - E_{\text{before}}(J)}{E_{\text{after}}(J) - E_{\text{before}}(J)}$$

Without the usual error ($b = 0$), $S(J) = 1 - \mathrm{OS}(J)$ exactly. For one jump, $(p_j - O_j)\, d_j$ is $p_j - O_j$ for an up jump and $(1 - p_j) - (1 - O_j)$ for a down jump, which is the implied rate minus the hit; and $|\Delta p_j|$ is the after-rate minus the before-rate. Summing over J and dividing by $|J|$ gives $\mathrm{OS} = (E_{\text{after}} - A)/(E_{\text{after}} - E_{\text{before}}) = 1 - S$. So accuracy is the overshoot seen from another angle, not a separate test.

Uncertainty

Ranges. Jumps of the same market are not independent, so whole markets are resampled: draw as many markets as there are, with replacement, M times; recompute the statistic each time; the range is the 2.5th to the 97.5th percentile of the M values (a 95% range).

Trader groups (F2)

Groups. A jump is made by "one trader" if $K_j = 1$, by "a crowd" if $K_j \geq K$, and by "a few traders" in between.

All factors at once. A weighted least-squares regression over jumps, with weight $|\Delta p_j|$:

$$\frac{o_j(p_j)}{|\Delta p_j|} = \beta_0 + \beta_{\text{few}}\, \mathbb{1}[1 < K_j < K] + \beta_{\text{crowd}}\, \mathbb{1}[K_j \geq K] + \sum_k \gamma_k x_{jk} + e_j$$

The x_{jk} are 0/1 indicators for question type, category, year, direction and, for each continuous control (money, price before the jump, time to the end date, market lifetime), its quantile bin among C equal-count bins. Weighting by $|\Delta p_j|$ makes the fitted values reproduce the overshoot shares, so β_{crowd} is the crowd-vs-one-trader gap with all the controls held fixed.

Wallets. For each wallet with at least m_w single-trader jumps, OS of the first half of its jumps (in time) and of the second half; the report gives their correlation across wallets. A high correlation would mean some wallets always overshoot.

Other candidates (F4)

Offset. For a jump j in a neg-risk race, with Δp_m the price change of each other candidate m in the same slot:

$$\phi_j = -\frac{\sum_{m \neq j} \Delta p_m}{\Delta p_j}$$

$\phi = 1$: the others fell by exactly as much as j rose; $\phi = 0$: they did not move. Groups: no other candidate traded; $\phi < \phi_{\text{lo}}$ (not offset); $\phi_{\text{lo}} \leq \phi < \phi_{\text{hi}}$ (partly); $\phi \geq \phi_{\text{hi}}$ (offset).

Betting against jumps (F5)

Return per dollar. Right after the jump, buy the opposite side at e_j, the VWAP of the first N trades on the bettor's side within T (after an up jump, buy NO at $1 - e_j$; after a down jump, buy YES at e_j), and hold to the end:

$$\rho_j = \frac{d_j\,(e_j - O_j)}{\sigma_j}, \qquad \sigma_j = \begin{cases} 1 - e_j & d_j = +1 \\ e_j & d_j = -1 \end{cases}$$

σ_j is the price of the bet (the stake per share) and $d_j(e_j - O_j)$ the gain per share. "Net" removes the usual error: $\rho_j^{\text{net}} = \rho_j + d_j\, b(e_j)/\sigma_j$.

Which bets, and the trimmed mean. Bets only on jumps that started inside $[\eta, 1-\eta]$; below that the opposite side costs a few cents and one win pays dozens of times the stake. The reported return clips every ρ_j^{net} to the τ and $1-\tau$ quantiles of all bets, then averages.

Spread. The median of $(e_j^{\text{with}} - e_j)\, d_j$, where e_j^{with} is the same average on the side of the jump: what crossing from one side to the other costs.

Market structure (F6)

Computed per market, over its used slots, for markets with at least $S_{\min}$ slots with trades.

- **One-trader rate (the target):** single-trader jumps per 100 slots with trades.
- **Kyle's lambda:** the slope λ of $\Delta\ell_s = \alpha + \lambda F_s + e_s$ fitted by least squares, where $\Delta\ell_s$ is the log-odds change from the previous slot (at most $g_{\max}$ apart) and F_s is the net taker buying of YES in the slot, in thousands of dollars. It is how far $1,000 of net buying moves the price.
- **Roll spread:** $2\sqrt{-\operatorname{cov}(\Delta y_i, \Delta y_{i-1})}$ over consecutive trades (0 if the covariance is positive). When trades bounce between the buy and the sell price, consecutive changes have negative covariance, and its size gives the spread.
- **Buy-sell gap:** in each window of w_s seconds with both taker buys and taker sells, the VWAP of the buys minus the VWAP of the sells; the market's median.
- **Variance ratio:** $\mathrm{VR}(L) = \operatorname{var}(\ell_s - \ell_{s-L})/(L \cdot \operatorname{var}(\ell_s - \ell_{s-1}))$. Below 1, changes tend to reverse.
- **Two-sidedness:** $2\sum_s \min(B_s, S_s)/\sum_s (B_s + S_s)$, with B_s, S_s the taker buys and sells in dollars; 1 = perfectly balanced, 0 = one-sided.
- **Concentration:** with a_w wallet w's share of the market's taker dollars, the Herfindahl index $H = \sum_w a_w^2$, the top wallet's share $\max_w a_w$, the share of the top 10% of wallets, and the Gini index $G = \frac{\sum_i \sum_k |x_i - x_k|}{2n^2\bar{x}}$ over the n wallets' dollar amounts x (0 = equal, 1 = one wallet).
- **Partial rank correlation:** each measure and the target are turned into ranks, the ranks are regressed on the log of traders, dollars and slots (market size), and the correlation of the two residuals is reported. It answers "at the same size, do markets with a higher measure see more single-trader jumps?"

Tags: the same, over tags with at least n_{tag} markets.

Crypto up/down (F8)

The same definitions, with the crypto values of Δ, θ, c_{start}, c_{end}, ε, κ, N, T, the crowd K, the bins W, the resamples M and the outcome rule Q (section 6.2). The usual error is measured at the prices before the crypto jumps, per coin, market length and price bin. The related market (the other coin, or the same coin's market of another length) "moved" in the jump's minute if its log-odds changed by at least μ over that minute.

Pattern search (F9)

Path. For a jump j, its prices at steps of a minutes from b_- minutes before to b_+ minutes after the slot's close, each the last slot price at that time, in log-odds: a window of $m = (b_- + b_+)/a + 1$ points. Paths need price changes in at least n_{chg} of their post-jump steps, and jumps need at least H hours of the market's window left.

Distance and matrix profile. Two windows X and Y of m points are compared after z-normalising each (subtracting its mean and dividing by its standard deviation), so only their shapes count:

$$D(X,Y) = \sqrt{2m\left(1 - \rho_{XY}\right)}$$

where ρ_{XY} is their correlation. The matrix profile of a window is its distance to its nearest neighbour in *another* event: $P_X = \min_Y D(X,Y)$.

Chance comparison. Surrogate paths keep each jump's pre-jump part and jump, with the post-jump steps shuffled. A path "has a close match" if $P_X < R$, where R is the r-th percentile of the surrogates' profiles. Real paths are compared with the surrogates at the same R; a recurring shape would give many more matches than chance. The search uses every path that passes the filters.

Crypto chains. Each up/down market's price at n_{cx} evenly spaced points over its window, for markets with at least n_{tr} trades in it; the markets of one coin back to back form one long series, searched with windows of one market (n_{cx} points).

Signs at the time of a jump (F10)

crowd: $K_j \geq K$; money: $U_j \geq X$; co-jump: another market of the same event jumped in the same slot; follow-up: a trade within τ_f seconds of the slot's close.

The elbow method

Knee of a curve. Given a curve (x_k, y_k), $k = 0, \ldots, n$, over a stated range, both axes are scaled to [0, 1]:

$$\widehat{x}_k = \frac{x_k - \min x}{\max x - \min x}, \qquad \widehat{y}_k = \frac{y_k - \min y}{\max y - \min y}$$

and the knee is the point farthest from the straight line through the first and the last point (the Kneedle rule, Satopää et al. 2011):

$$k^* = \operatorname*{argmax}_k \frac{\left|(\widehat{y}_n - \widehat{y}_0)(\widehat{x}_k - \widehat{x}_0) - (\widehat{x}_n - \widehat{x}_0)(\widehat{y}_k - \widehat{y}_0)\right|}{\sqrt{(\widehat{x}_n - \widehat{x}_0)^2 + (\widehat{y}_n - \widehat{y}_0)^2}}$$

Before the knee the curve changes fast; after it, slowly. The axis of x is always linear, even for parameters tried over a wide range (dollars, seconds), where some figures display it on a log scale for reading only. For curves that are shares of jumps, only points backed by at least 1,000 jumps are used.

Parameters that depend on each other are set together by a fixed-point search. Start from any values; set each parameter to the knee of its curve with all the others at their current values; repeat until no knee moves:

$$\theta^{(t+1)} = \text{knee}_\theta\left(\Delta^{(t)}, c^{(t)}, \ldots\right), \quad \Delta^{(t+1)} = \text{knee}_\Delta\left(\theta^{(t+1)}, c^{(t)}, \ldots\right), \quad \ldots$$

At the end, every parameter is the knee of its own curve given all the others.

6.2 Parameters and how each was chosen

Every value is the knee of a curve (Kneedle rule, linear axis), drawn in section 2, on every market. "Jumps kept" curves count the price jumps; the jump-definition knees were found together, each with the others at their knees.

What a price jump is (event markets)

Parameter	Symbol	Value	Curve	Range tried
Slot length	Δ	**30 min**	jumps kept	1 min - 2 h
Jump threshold	θ	**15 points**	jumps kept	5 - 50 points
Cut at the start of life	c_{start}	**7%**	jumps kept	0 - 30%
Cut at the end of life	c_{end}	**4.5%**	jumps kept	0 - 30%
Settle band	ε	**3 points** (3-97%)	jumps kept	1 - 20 points
Price clip	κ	**2 points** (nearly flat curve)	jumps kept	0.1 - 5 points
Next trades: how many	N	**5 trades**	overshoot at the next trades	1 - 16
Next trades: how long	T	**45 min**	jumps with next trades	1 - 180 min
Outcome	Q	**0.995**	markets with a clear outcome	0.5 - 1

Crypto up/down markets (F8)

Parameter	Value	Range tried
Slot length	**1 min**	15 s - 5 min
Jump threshold	**18 points**	5 - 50 points
Cut at the start / end of life	**18% / 7.5%** (the start cut's curve is nearly flat)	0 - 30%
Settle band, price clip	**3 points, 3 points**	1 - 20, 0.1 - 5 points
Next trades	**5 trades within 2 min**	1 - 16 trades, 15 s - 15 min
A related market "moved" (μ)	**0.75 in log-odds**	0.01 - 1
Crowd	**6 traders** (groups 1 / 2-5 / 6+)	1 - 30
Price bins of the baseline	**8**	2 - 100
Resamples for the ranges	**150**	10 - 1,000
Outcome	**0.6** (prices settle at exactly 0, 1 or a split)	0.5 - 1

Statistics and groups

Parameter	Symbol	Value	Curve	Range tried
Random sample for the baseline	ρ	**1% of slots**	spread of the overshoot over 20 redraws	0.1 - 5%

Parameter	Symbol	Value	Curve	Range tried
Price bins of the baseline	W	**20** (5 points each)	share of the calibration error explained	2 - 100
Resamples for the ranges	M	**150**	spread of a range's end over 40 repeats, fitted as $c/\sqrt{M}$	10 - 1,000
Crowd	K	**5 traders**	overshoot of jumps with at least K traders	1 - 30
Big money	X	**$200**	overshoot of jumps with at least X in the slot	$1 - $100k

Settings of the findings

Finding	Parameter	Symbol	Value	Curve	Range tried
F1	tags shown		**200+ markets with jumps**	tags kept	10 - 2,000
F2	wallets in the persistence check	m_w	**8+ single-trader jumps**	wallets kept	2 - 50
F2	bins of each continuous control	C	**3**	crowd gap with all else fixed	2 - 10
F4	"not offset" below / "offset" from	ϕ_{lo}, ϕ_{hi}	**0.55 / 1.0**	crowd overshoot below / from the cut	0.05 - 1.5
F5	price band of the bets	η	**none** (flat curve)	spread of the returns	0 - 20 points
F5	trim at each end	τ	**2.5%**	trimmed mean return	0 - 10%
F6	slots per market	$S_{\min}$	**50+**	markets kept	5 - 200
F6	Kyle's lambda: slots at most apart	$g_{\max}$	**8 h**	pairs of slots kept	0.5 - 48 h
F6	variance ratio lag	L	**6 slots**	median variance ratio	2 - 16
F6	buy-sell window	w_s	**120 s**	windows with both sides	10 s - 10 min
F6	tags	n_{tag}	**200+ markets**	tags kept	10 - 2,000
F9	step between path points	a	**10 min**	error against the 1-minute path	1 - 30 min
F9	path start before / end after the close	b_-, b_+	**30 / 120 min**	distance from the price at the close	0 - 60, 0 - 480 min
F9	price changes in the path	n_{chg}	**3+**	paths kept	0 - 12
F9	hours of window left	H	**2+**	paths that run into the settle band	0 - 24 h
F9	match radius	r	**15th percentile**	surrogate profile	1 - 50
F9	points per crypto market	n_{cx}	**30**	error against a 120-point path	4 - 120
F9	trades per crypto market	n_{tr}	**50+**	markets kept	1 - 100
F10	follow-up trade within	τ_f	**20 min**	overshoot of jumps with a follow-up	1 - 45 min

Not tuned: the 95% level of the ranges (a convention); the splits used only to display a finding (price bands, time to the end date, years, sports, histogram bins).

6.3 Pattern search (F9)

Search	Series	Window	Distance	Chance comparison
After price jumps	16 prices per jump, every 10 min from 30 min before to 120 min after the slot's close; 25,355 jumps with at least 3 price changes and 2 hours of window left	16	z-normalised; matches must be in another event	post-jump steps shuffled; radius = 15th percentile of the shuffled paths' profile
Bitcoin chain	bitcoin 15-min markets back to back, 30 points each, markets with 50+ trades	30	z-normalised, `stumpy.stump`	profile vs shuffles
Coin pairs	the 4 coins' 15-min chains, AB-joins	30	`stumpy.stump(T_A, 30, T_B)`	other coin shuffled in time

6.4 Computing and plots

- **Server:** each extraction step runs on 2 threads within 8 GB of memory. Regrouping the 766 million trades takes one pass of about 30 minutes.
- **Plots:** Python with numpy, pandas, scipy, stumpy and matplotlib, in the Rose Pine Dawn theme. The trader groups have fixed colours: 1 trader = rose-red, 2-4 = gold, 5+ = pine blue.

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7. Glossary

Term	Meaning
YES price	Price of the share that pays $1 if the event happens; read as a probability
Log-odds	`log(p / (1 - p))`; stretches the scale near 0% and 100%
Slot	A 30-minute window in which trades are grouped
Price jump	A slot whose price is ≥ 15 points from the previous slot with trades
Life of a market	From its first trade to the moment its price leaves 3-97% for good (or its last trade); the first 7% and the last 4.5% are left out
Maker / taker	The maker posted an order and waited; the taker accepted it, moving the price ("taking liquidity")
Overshoot	The part of a jump that the final outcome did not justify, beyond the usual error, as a % of the jump
Next trades	The first five trades after a jump's slot, within 45 minutes
Neg-risk event	A Polymarket race whose candidates exclude each other; a NO share in one converts into YES shares of all the others, which links their prices and gives the long shots liquidity
Spread	The gap between the price to buy and the price to sell
Kyle's lambda	How far $1,000 of net buying moves the price
Gini / Herfindahl	Measures of how concentrated trading is among wallets
Matrix profile	For every window of a series, the distance to its most similar other window

8. Where the code is

Step	Scripts
Server extraction	`remote/s1_bucket.py` (regroup trades), `s2_events.py` / `s3_events.py` (slots, window and jumps), `s4_paths.py` (price paths, crypto jumps), `s6_micro.py` (market structure), `s7_events.py` (plain averages), `s8_wallets.py` (wallets and other candidates), `p7_extract.py` (example market)
F1-F3	`explore/r02_persistence.py`, `r03_robustness.py`, `r04_types.py`, `r07_basis.py`
F2 regression	`explore/r05_event_markets.py`
F4	`explore/r08_pairs.py`, `r07_siblings.py`
F5, F6	`explore/r06_fade.py`, `r08_fade_robust.py`, `r06_concentration.py`
F7, F8	`explore/r07_sports.py`, `r05_updown.py`
F9	`explore/r04_motifs.py`, `r04_btc_chain.py`, `r05_crypto_mp.py`
F10	`explore/r10_information.py`
Parameters (elbow)	`remote/s12_grid.py`; `explore/elbow.py` (the knee), `r12_joint.py`, `r12_params.py`, `r12_f9.py`; every value in `explore/r02_persistence.py`
Traders one by one, give-back, pointing the right way	`explore/r11_traders.py`
This report	`explore/report_stats.py`, `report_figures.py`, `report_style.py`

Related work: Tsang & Yang, "Political Shocks and Price Discovery in Prediction Markets" and "The Anatomy of Polymarket" (arXiv 2603.03152, 2603.03136); Akey, Vincent, Harvie & Martineau, "Who Wins and Who Loses in Prediction Markets? Evidence from Polymarket" (CEPR DP 21615, 2026); Roll (1984).